A NOVEL APPROACH TO MODEL THE PION STRUCTURE USING ADVANCED POLYNOMIAL FUNCTIONS

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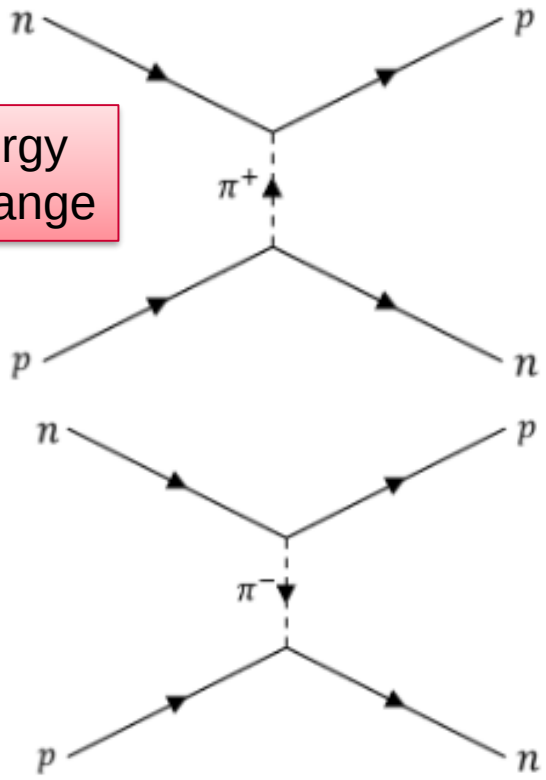
Contents of This Talk

1. Central role of the pion within QCD.
2. Overcoming difficulties in studying the pion structure in scattering experiments.
3. The impact of modeling assumptions on pion parton distribution functions (PDFs).
4. Advantages of advanced polynomial parameterizations of PDFs for studying these questions.

Precision studies of the pion structure

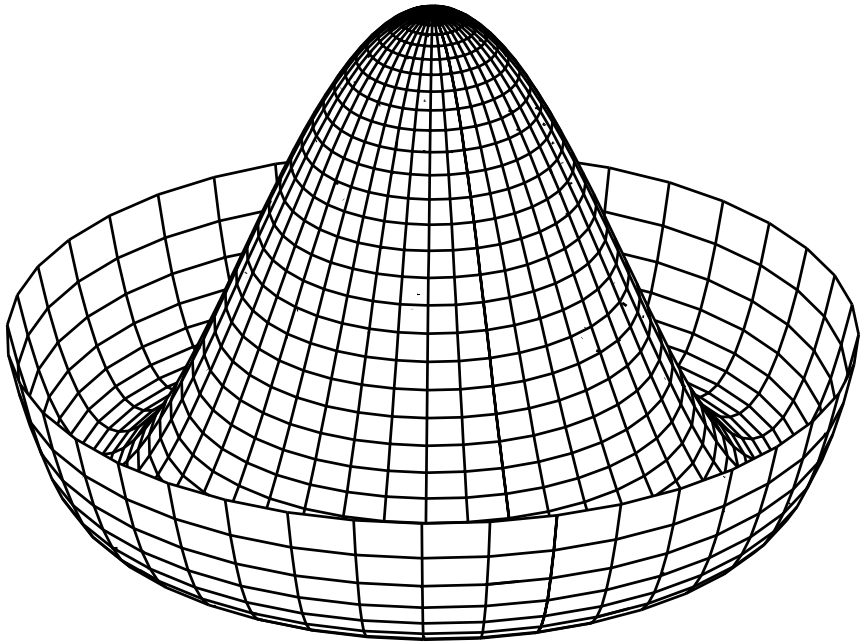
The Pion in Nuclear and High-Energy Physics

Low energy pn exchange



- Unstable strongly interacting particle with $m_{\pi^\pm} = 140\ \mathrm{MeV}$.
- Mediates the strong force between nucleons at low energy.
- Central to chiral symmetry breaking and low-energy QCD.
- Lightest hadron; pseudo-Nambu-Goldstone boson (NGBs).

Chiral Symmetry Breaking (χSB)

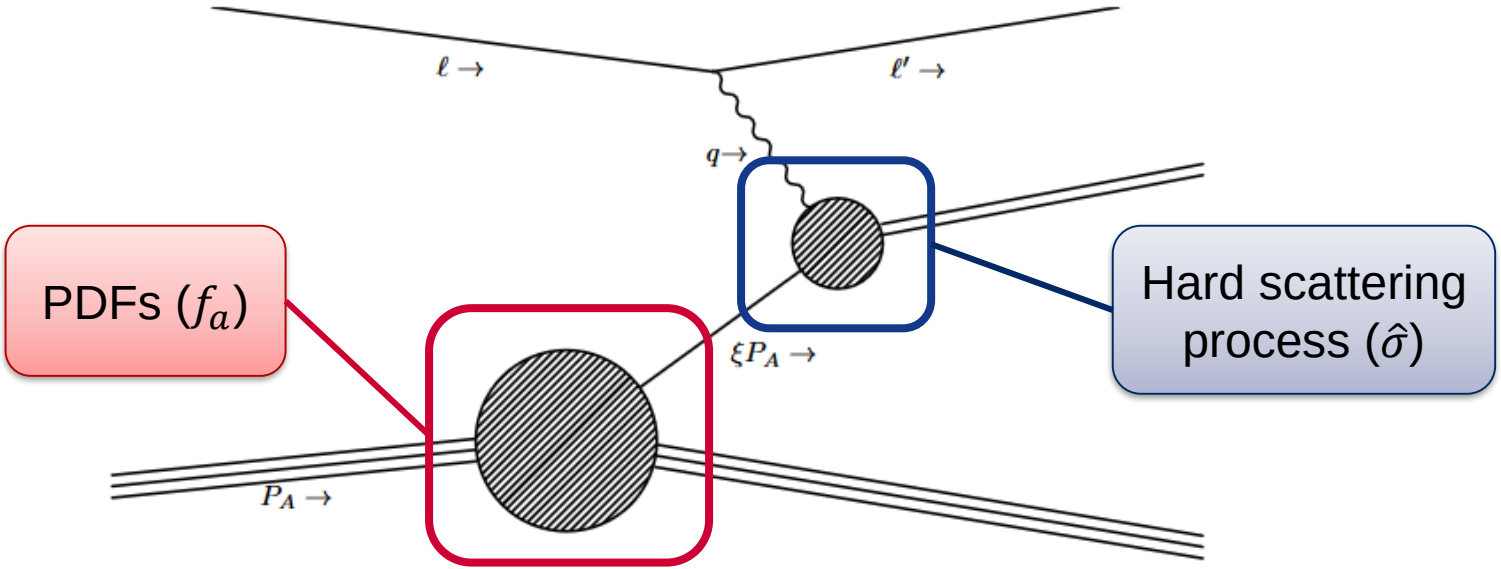


- QCD with massless quarks results in *spontaneously broken* chiral symmetry.
- Electroweak symmetry breaking (Higgs mechanism) does not account for all visible mass; χSB does.
- The $\mathrm{SU}(2)$ axial symmetry breaking, $\mathrm{SU}(2)_{\mathrm{L}} \times \mathrm{SU}(2)_{\mathrm{R}} \rightarrow \mathrm{SU}(2)_{\mathrm{V}}$ isospin symmetry breaking predicts 3 massless bosons: π^0, π^+, π^-.
- Pions gain small mass when $m_u \neq m_d \neq 0$, *pseudo*-NGBs.

Why study the structure of mesons?

- Simplest hadronic bound states, composed of constituent quark-antiquark pair surrounded by the quark-gluon sea.
- By studying the structure of pions, kaons, nucleons, and other hadrons, we develop theoretical methods for modelling hadrons in non-perturbative QCD (Lattice QCD, low energy χSB models, Dyson-Schwinger equations,…).
- **Parton distribution functions ($f_a(\xi_a, \mu_F)$) quantify the structure of hadrons in inelastic scattering processes.**

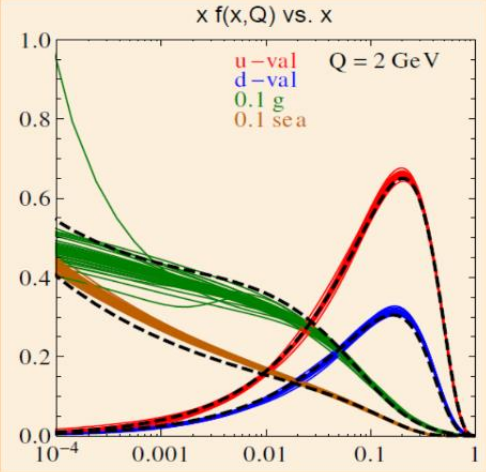
Factorization of QCD Cross Sections (Ex: Lepton-hadron Deep-Inelastic Scattering)



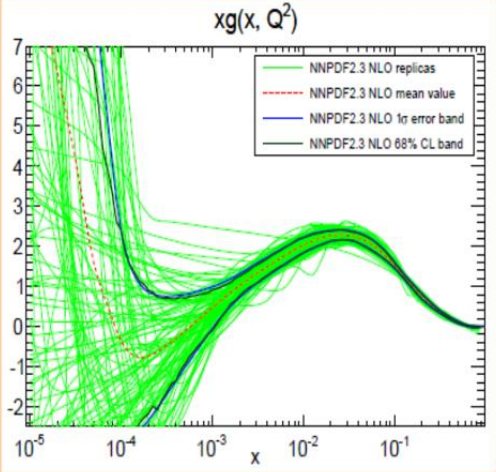
$$\sigma_{\mathrm{DIS}}(Q) = \sum_{a=g,q,\bar{q}} \int_0^1 d\xi_a \, \hat{\sigma}\left(\frac{x_a}{\xi_a}, \frac{Q}{\mu_R}, \frac{Q}{\mu_F}; \alpha_s(\mu_R)\right) \times f_a(\xi_a, \mu_F) + \mathcal{O}\left(\frac{\Lambda_{\mathrm{QCD}}^2}{Q^2}\right)$$

Two Representations of PDFs

Analytic parametrizations + Hessian PDF eigenvector sets **(ABM, HERA, CTEQ, MMHT,…)**



Neural network parameterizations + Monte Carlo PDF replicas **(NNPDF)**



$$f_a(\xi_a = x, \mu_F^2 = Q^2)$$

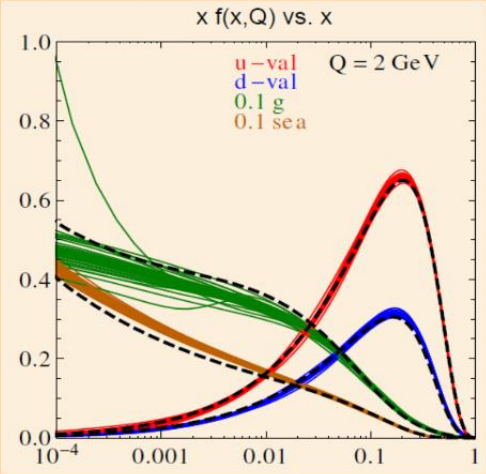
Parton distribution functions (PDFs) encode **nonperturbative** probabilities for finding partons in the hadron.

Scale dependence from DGLAP evolution.

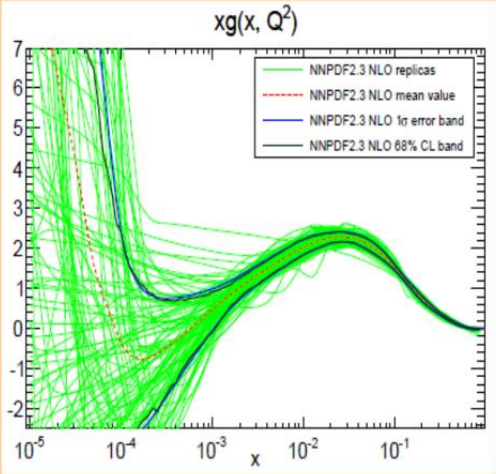
Usually determined from experimental data.

Two Representations of PDFs

Analytic parametrizations +
Hessian PDF eigenvector sets
(ABM, HERA, CTEQ, MMHT,…)



Neural network parameterizations
+ Monte Carlo PDF replicas
(NNPDF)



<u>Analytic parameterizations</u>:

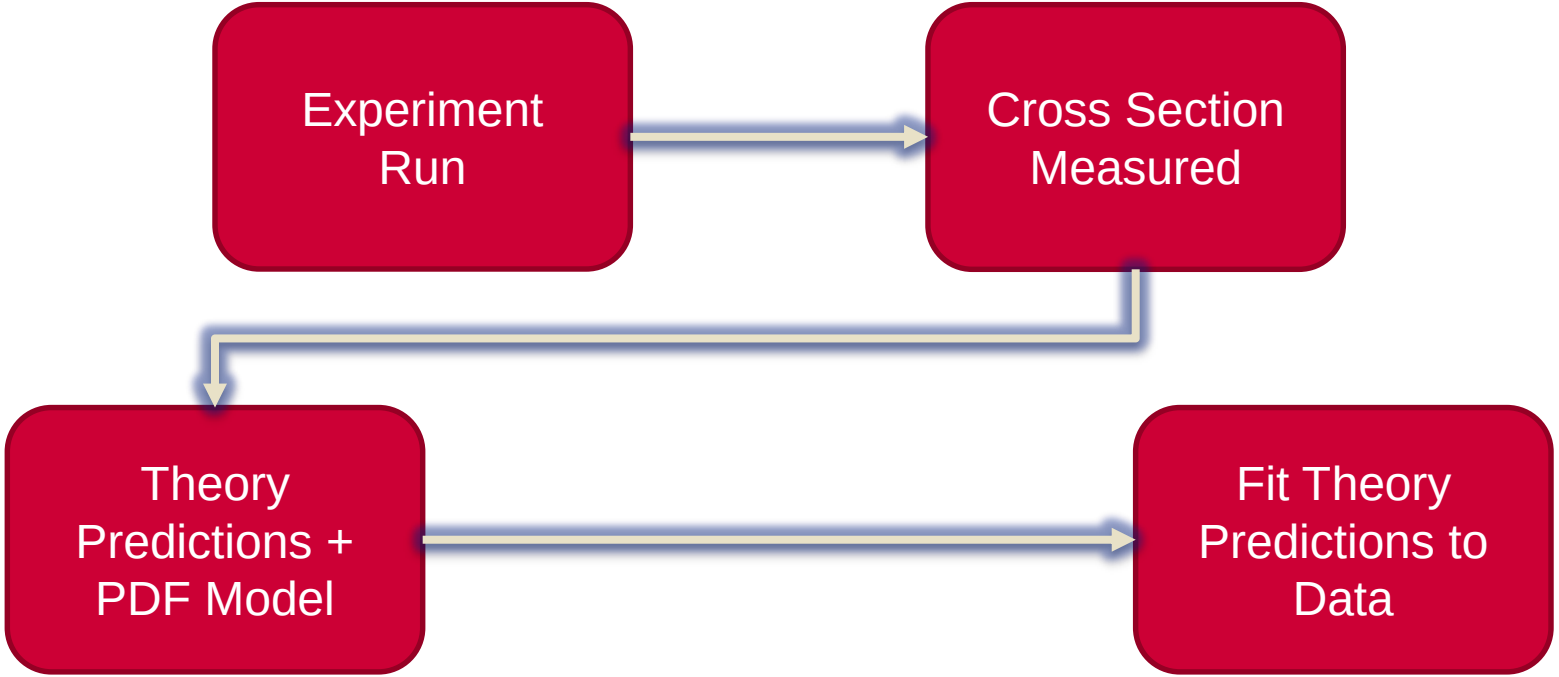
Polynomials – controllable, less flexible.

<u>Neural network (NN) parameterizations</u>:

Universal approximators, less control.

NEW: Advanced analytic parameterizations that can outperform NN ones.

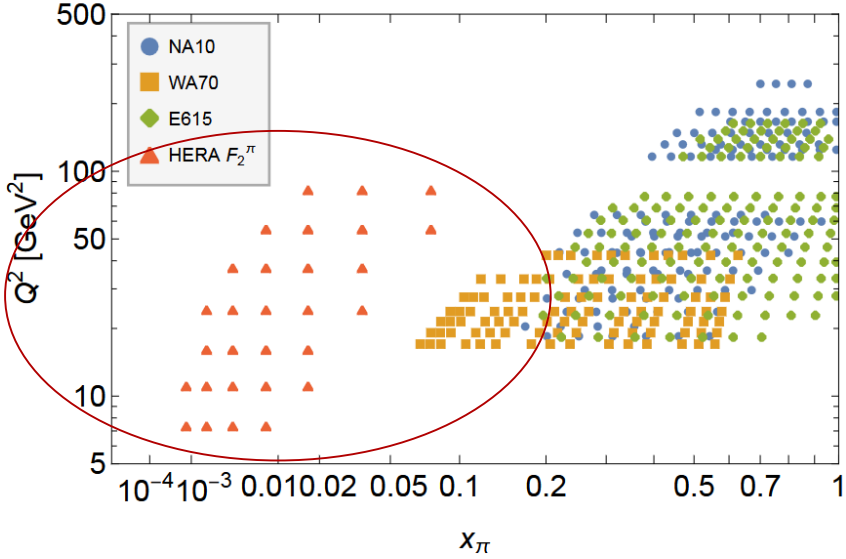
PDFs are determined using global fits of experimental data.



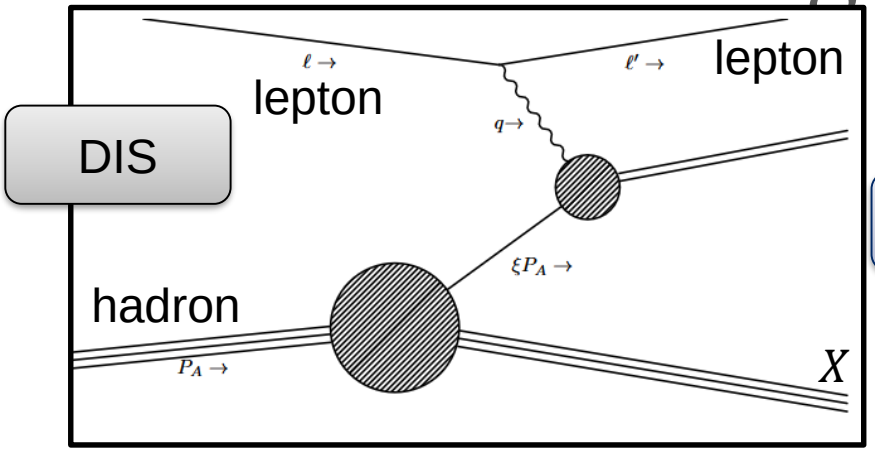
Pion scattering data are still limited.
More expected soon at the Electron-Ion Collider

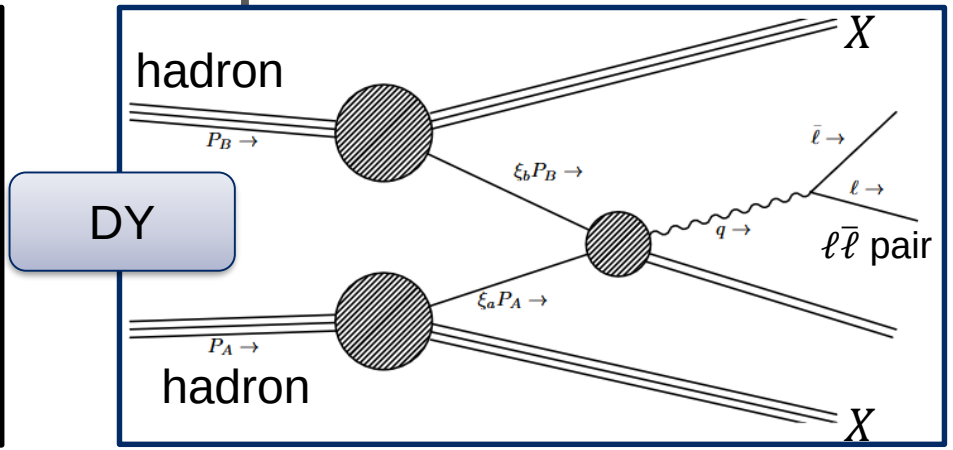
Pion datasets in our 2025 study

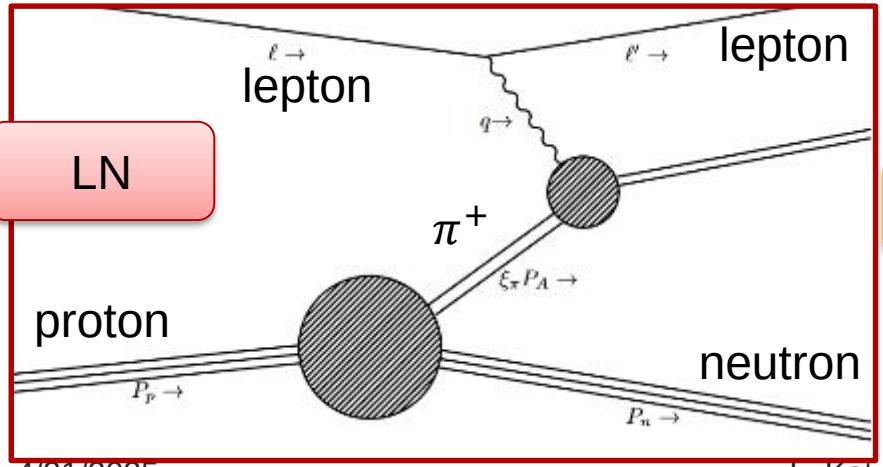
- Drell-Yan (DY) (**E615** and **NA10**)
  - constrains valence at large x.
- Prompt-photon production (**WA70**)
  - constrains gluon at intermediate x.
- Pion DIS from leading-neutron processes (LN) (**HERA**)
  - disentangle $S(x)$ and $g(x)$ and $x < 0.1$.
- Advanced polynomial parameterizations are critically important to explore PDF solutions allowed by data.

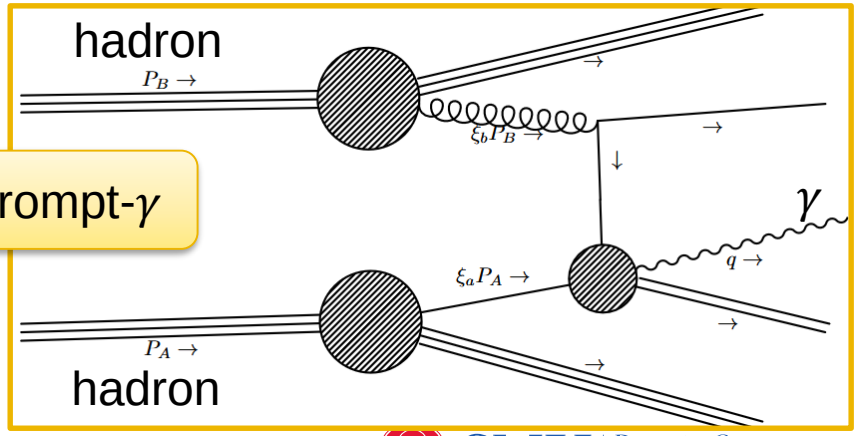


Processes in the global pion fit

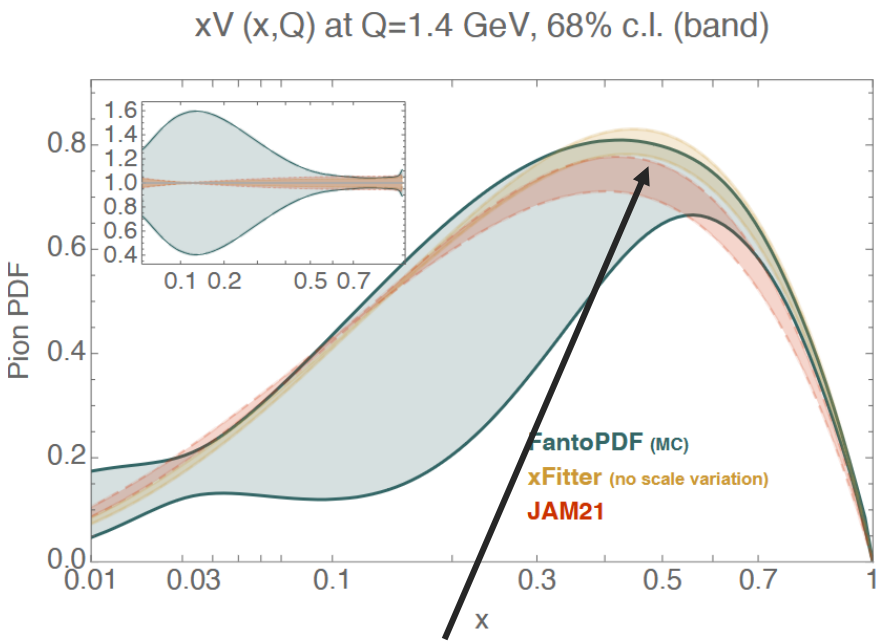








Efficient parameterization of PDFs is central for uncertainty quantification.



Parameterization dependence not included.

- **Aleatory (Stochastic)**: Random uncertainty from data fluctuations (e.g., statistical experimental errors)
- **Epistemic**: Uncertainty from modeling assumptions (e.g., systematic errors or parameterization uncertainties).

Parameterizing pion PDFs

- The assumptions made yield the following quark PDF relations:
  - $u^{\pi^+} = \bar{d}^{\pi^+} = \bar{u}^{\pi^-} = d^{\pi^-}$
  - $\bar{u}^{\pi^+} = d^{\pi^+} = u^{\pi^-} = \bar{d}^{\pi^-} = s^{\pi^\pm} = \bar{s}^{\pi^\pm}$
- Using these relations the **three** independent π^+ PDFs are:
  - Valence: $\boldsymbol{V(x) = [u(x) - \bar{u}(x)] + [\bar{d}(x) - d(x)] = 2[u(x) - d(x)]}$
  - Quark sea: $\boldsymbol{S(x) = 2d(x) + 2\bar{u}(x) + s(x) + \bar{s}(x) = 6d(x)}$
  - Gluon: $\boldsymbol{g(x)}$

Why Parameterization Matters

- Fixed functional forms can limit our results.
- We want to have a polynomial that is able to approximate any function while having control over the parameterization.
- Neural Networks (NNs) are universal approximators, but the fitting process is contained within a black box. The parameters are not physically interpretable.
- Our choice: Bézier curves – **universal approximators, interpretable, simple.**

PDF parameterizations using Bézier curves

Including a toy demonstration

Metamorph = Carrier × Modulator

$$f_{meta}(x) = f_{Carrier}(x) \times f_{modulator}(x)$$

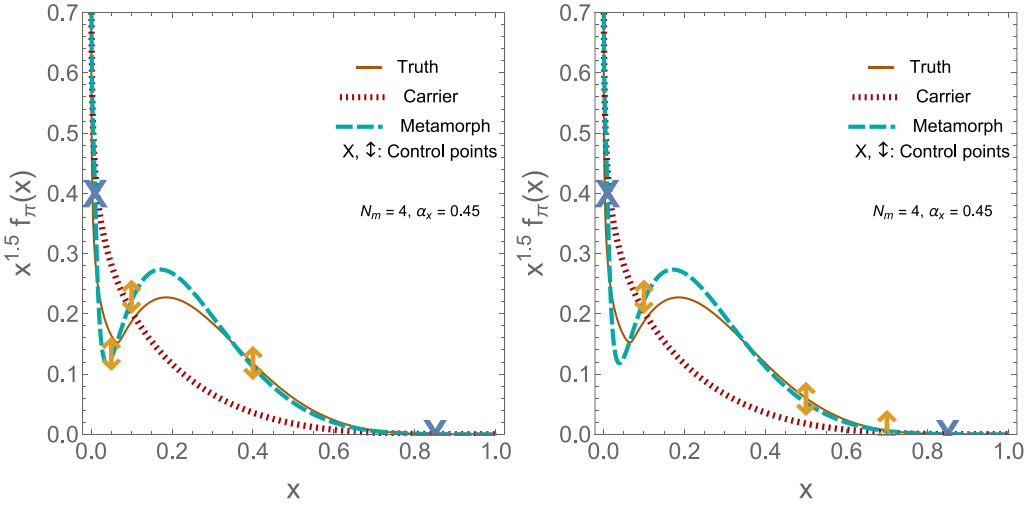
$$f_{Carrier}(x) = A_f x^{B_f}(1 - x)^{C_f}$$

$$f_{modulator}(x) = 1 + \mathcal{B}^{(N_m)}(x^{\alpha_x})$$

Physical conditions on PDFs

- PDFs have several conditions that must be satisfied.
  - $\int_0^1 f(x)dx = 1\,; \quad \lim_{x\to 0} f(x) = 0\,; \quad \lim_{x\to 1} f(x) = 0$
- To ensure the PDFs are physical, we enforce two sum rules:
  - $\langle v \rangle = \int_0^1 v(x)dx = 2$
  - $\sum_{f=v,S,g} \langle xf \rangle = 1$

$\mathcal{B}^{(N_m)}(x^{\alpha_x})$ are Bézier curves



Toy example

- Paul de Casteljau and Pierre Bézier first used them to design the Citroën DS with CAD.
- Defined by control points.
- Ideal for smooth, stable interpolation.

Bézier Curves

$$\mathcal{B}^{(n)}(x) \equiv \sum_{l=0}^{n} c_l B_{n,l}(x)$$

$$B_{n,l}(x) \equiv \binom{n}{l} x^l (1-x)^{n-l}$$

$$\Rightarrow \mathcal{B} = \boldsymbol{T} \cdot \boldsymbol{M} \cdot \boldsymbol{C}$$

$$\text{or } \boldsymbol{C} = \boldsymbol{M}^{-1} \cdot \boldsymbol{T}^{-1} \cdot \boldsymbol{P}$$

- $\mathcal{B}^{(N_m)}(y)$: Bézier function of N_m^{th}-degree.
- $\boldsymbol{C}$: $N_m + 1$ vector containing Bézier coefficients.
- $B_{N_m,l}(y)$: Bernstein basis polynomial.
- $\boldsymbol{M}$: A fixed $N_m + 1 \times N_m + 1$ matrix containing binomial coefficients. Determined by N_m.
- $\boldsymbol{T}$: A fixed $N_m + 1 \times N_m + 1$ matrix. Determined by the positions of control points.
- $\boldsymbol{P}$: $N_m + 1$ vector containing the values at the control points.

Bézier Curves

$$\mathcal{B}^{(n)}(x) \equiv \sum_{l=0}^{n} c_l B_{n,l}(x)$$

$$B_{n,l}(x) \equiv \binom{n}{l} x^l (1-x)^{n-l}$$

$$\Rightarrow \mathcal{B} = \boldsymbol{T} \cdot \boldsymbol{M} \cdot \boldsymbol{C}$$

$$\text{or } \boldsymbol{C} = \boldsymbol{M}^{-1} \cdot \boldsymbol{T}^{-1} \cdot \boldsymbol{P}$$

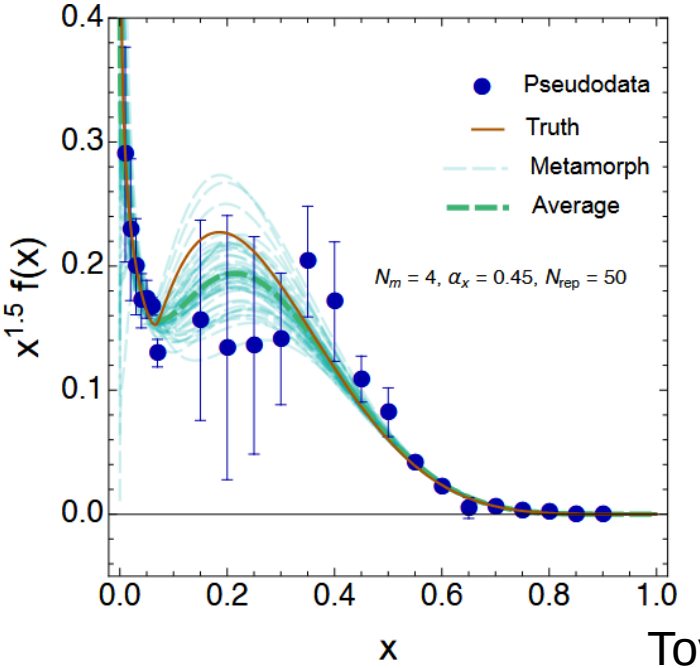
- Capable of *polynomial mimicry* – approximate any arbitrary function via the Weierstrass approximation theorem.
- Parameterizes the function values, P_i ⇒ easy physical interpretation of free parameters.
- Interpolates a smooth, continuous function from defined control points.

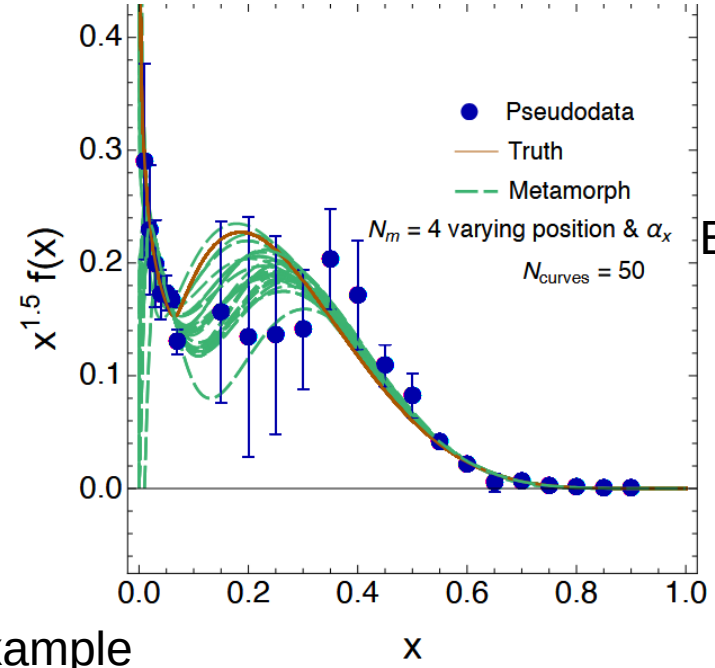
Bézier-based functions sample data fluctuations and parameterization variations

Bootstrap sampling
(fits fixed configuration to varied data)

Changing CPs and α_x ($y = x^{\alpha_x}$)
(fits varied functions to fixed data)

Aleatory



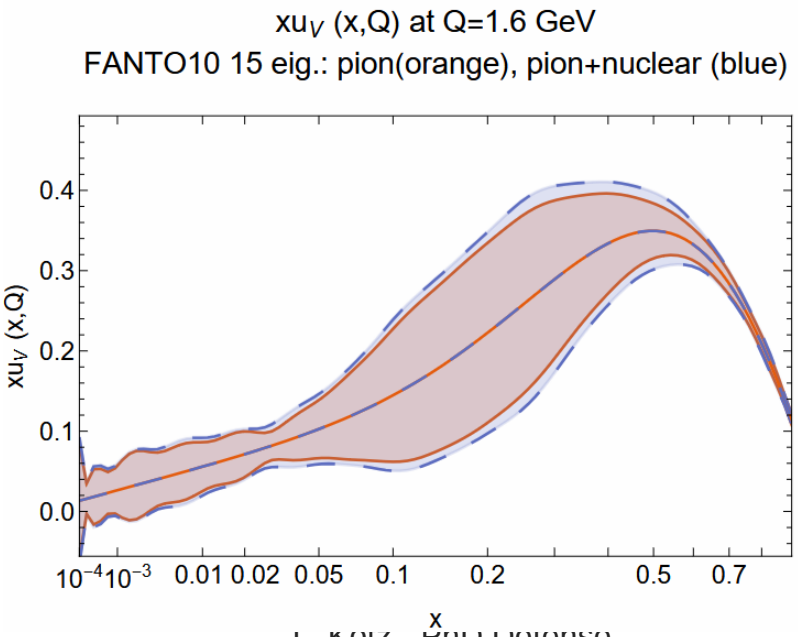


Epistemic

Toy example

FantoPDF 1.0: NLO pion PDFs with parameterization-driven and nuclear uncertainties (2311.08447) (2 papers in preparation)

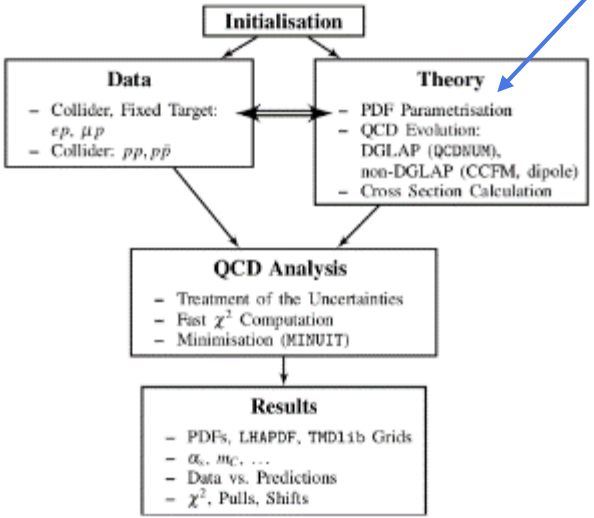
NEW



LHAPDF format
15+1 Hessian error sets,
Combining 5 independent parameterizations

Implementing Bézier curves into **xFitter**

```
lkotz@m3login02:~/mp4test/Fantomas4xFitter/pdfparams/Fantomas$ ls
adxmoment.h      fantomas.cc      Fantomas_PdfParam.cc   isNumber.h              metamorph.h
cl2DArray.h      fantomaschi2.h   Fantomas_PdfParam.h    LUPinverse.h
CMakeLists.txt   fantomas.h       integral.h             MetamorphCollection.h
```

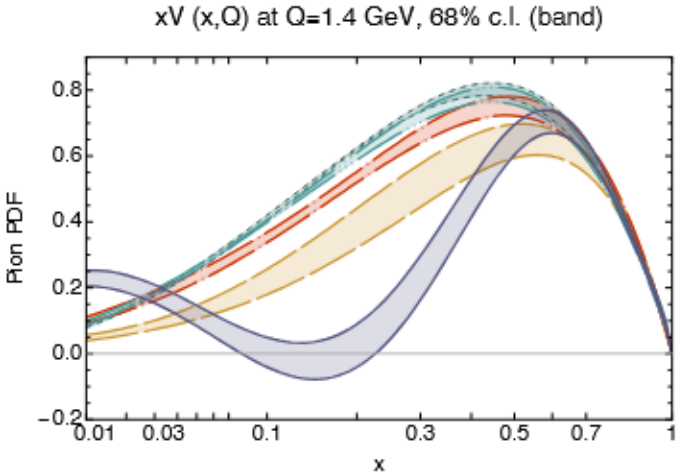


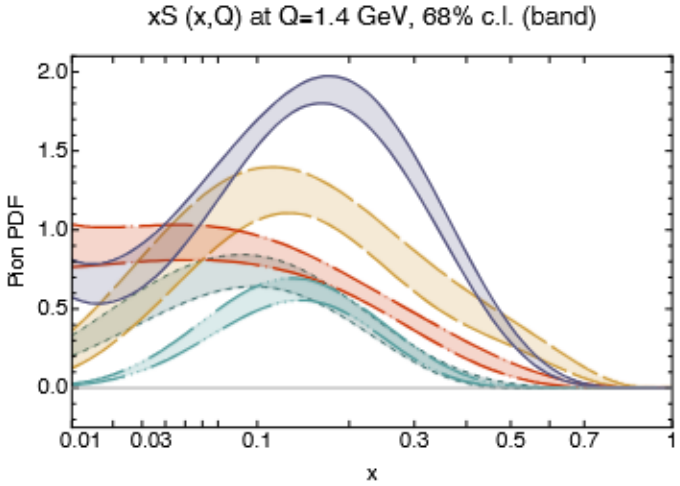
- **xFitter** is an open-source QCD fitting framework designed to extract PDFs from experimental data.
- Theoretical predictions are compared against experimental data via the χ^2 goodness-of-fit test.
- The Fantômas module was designed as a C++ package for **xFitter.**
- Fantômas contains metamorphs with specified input parameters, a collection of metamorphs with a user-defined ID, and functions allowing any external program to use metamorphs.

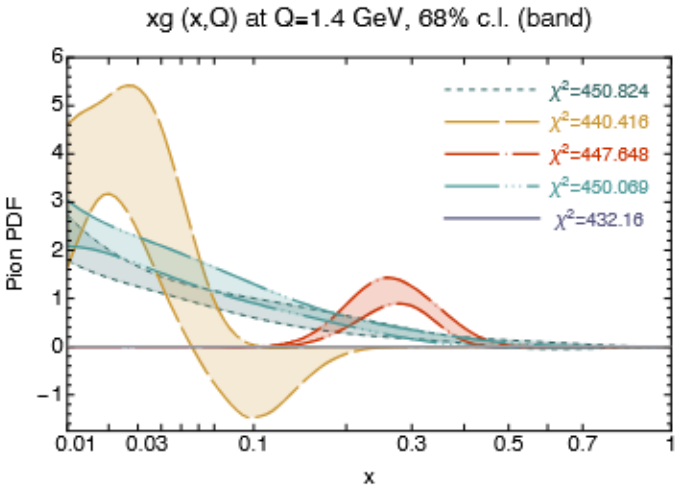
Assumptions made for pion PDFs

- PDFs are determined at next-to-leading (NLO) order in α_s.
- Corrections from electroweak theory and quark masses are neglected.
- Initial PDF scale set to $Q_0^2 = 1.9\ \mathrm{GeV}^2 < m_c^2$. Only light quarks (u, d, s) are present at the initial scale.
- The charm and bottom PDFs arise at higher scales from DGLAP evolution.

Selected Pion Fits (w/ LN)





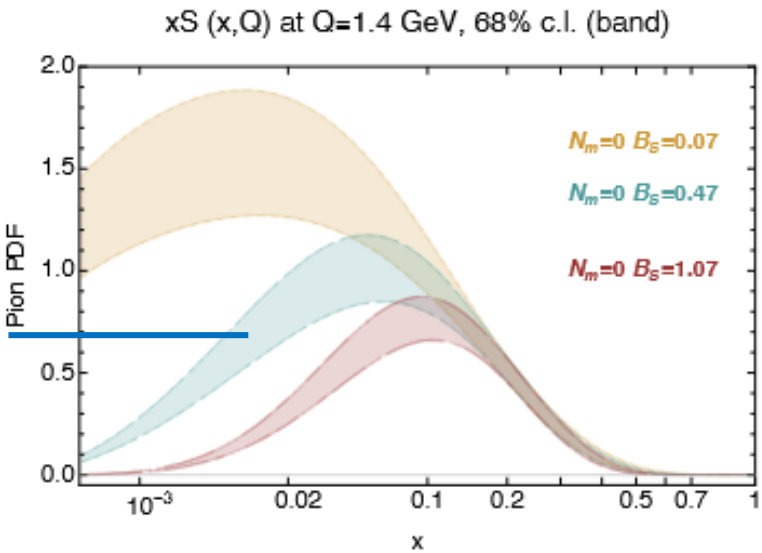


- 100 configurations were explored.
- 5 fits were selected with good χ^2 $(1.06 < \frac{\chi^2}{N_{pts}} < 1.10)$.
- Each selected fit contains an uncertainty band with $\Delta\chi^2 = 1$ (68% C.L.).

Comparing **xFitter** pion PDFs with Fantômas results (w/o LN data)

$$xf_i^{(0)}(x, Q^2) = A_i x^{B_i}(1-x)^{C_i}$$

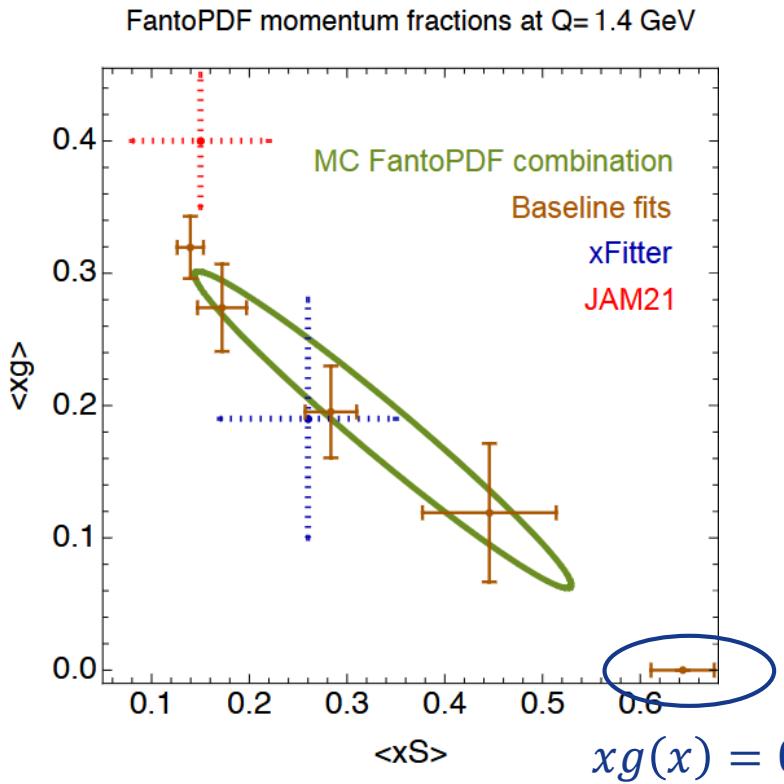
xFitter



$N_m = 0$ (DY+γ)	χ^2 [$d.o.f = 379 - 5$]
$B_S = 0.07$	445.70
$B_S = 0.27$	445.38
$B_S = 0.47$	445.29
$B_S = 0.67$	445.36
$B_S = 0.87$	445.52
$B_S = 1.27$	445.76

- The flexible Fantômas fits find more acceptable PDFs with good χ^2 than previous **JAM** and **xFitter** fits.
- The sea and gluon momentum fractions can invert and still produce statically similar results.

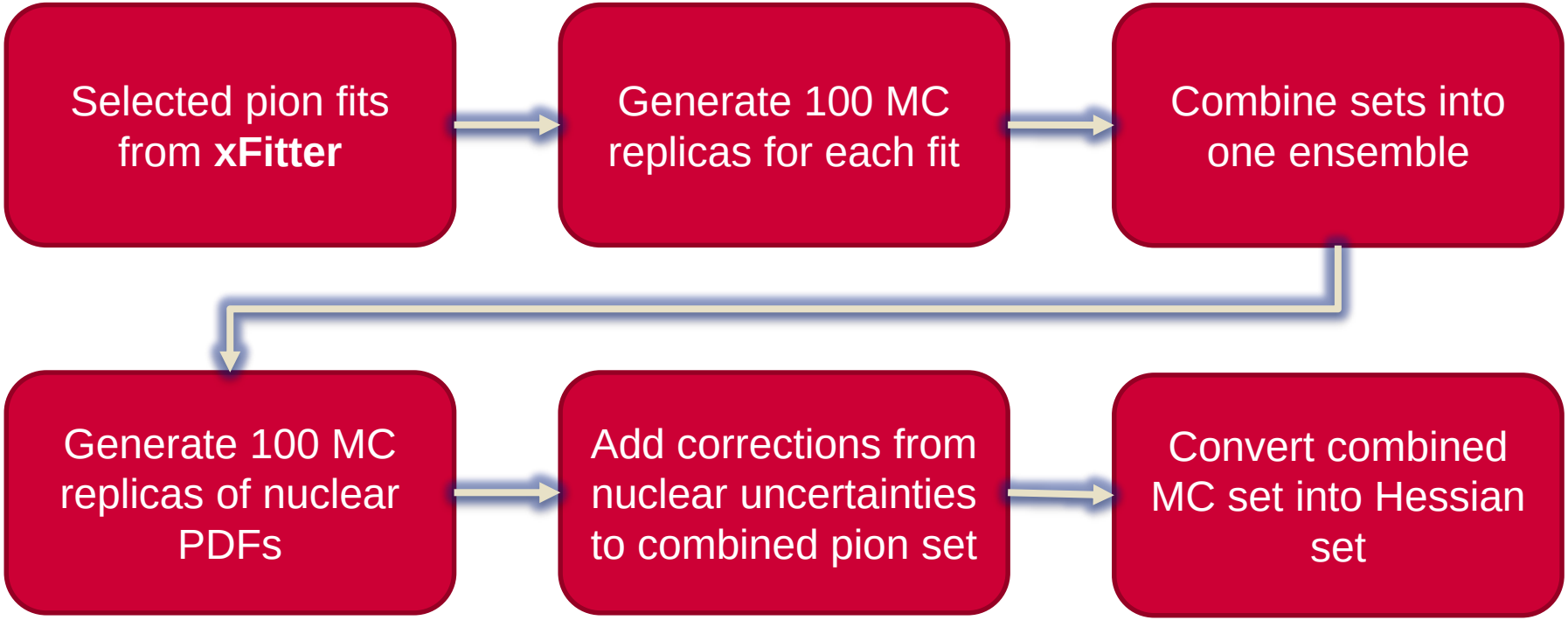
Pion Momentum Fractions of Selected Fits



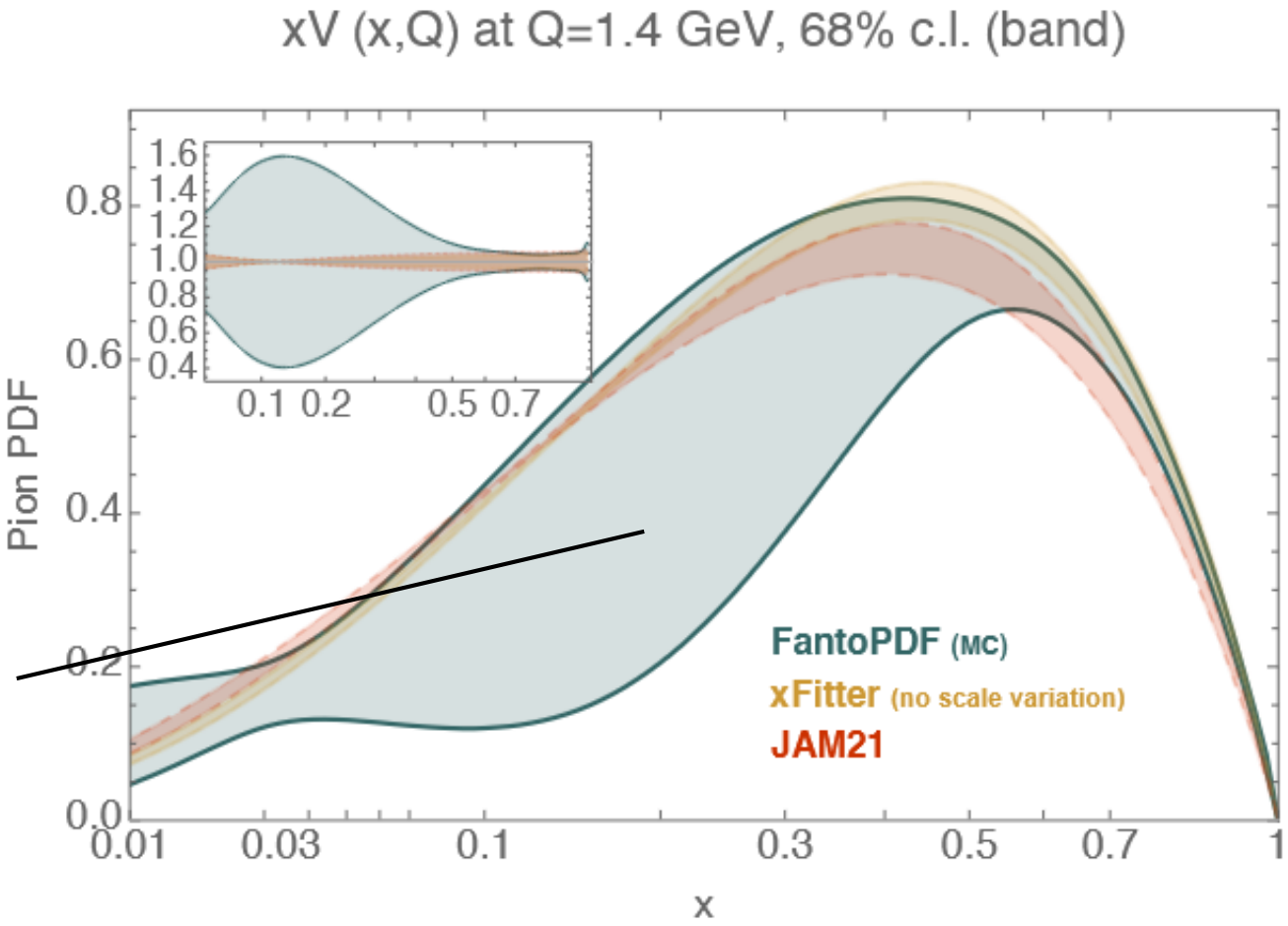
$xg(x) = 0$

- We calculate the uncertainty of the average momentum fraction, $\langle xf \rangle$, for each flavor.
- $\langle xV \rangle$ is approximately the same for all 5 fits.
- $\langle xS \rangle$ and $\langle xg \rangle$ contains a linear trend between each other.
- The lack of g and S constraints at low x results in degeneracy.

Quantifying the Epistemic Uncertainty

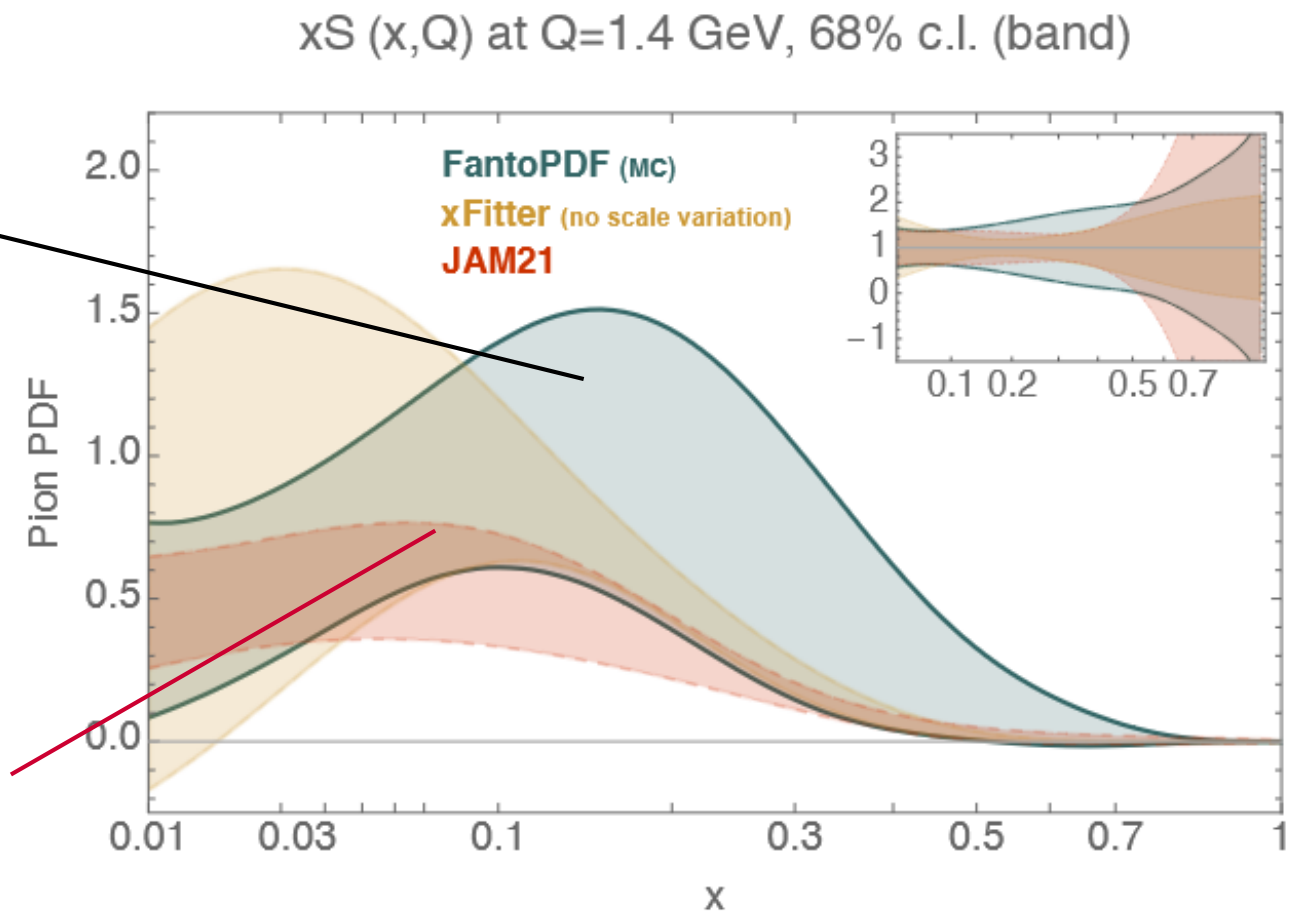


Inclusion of model-dependency result in larger uncertainties



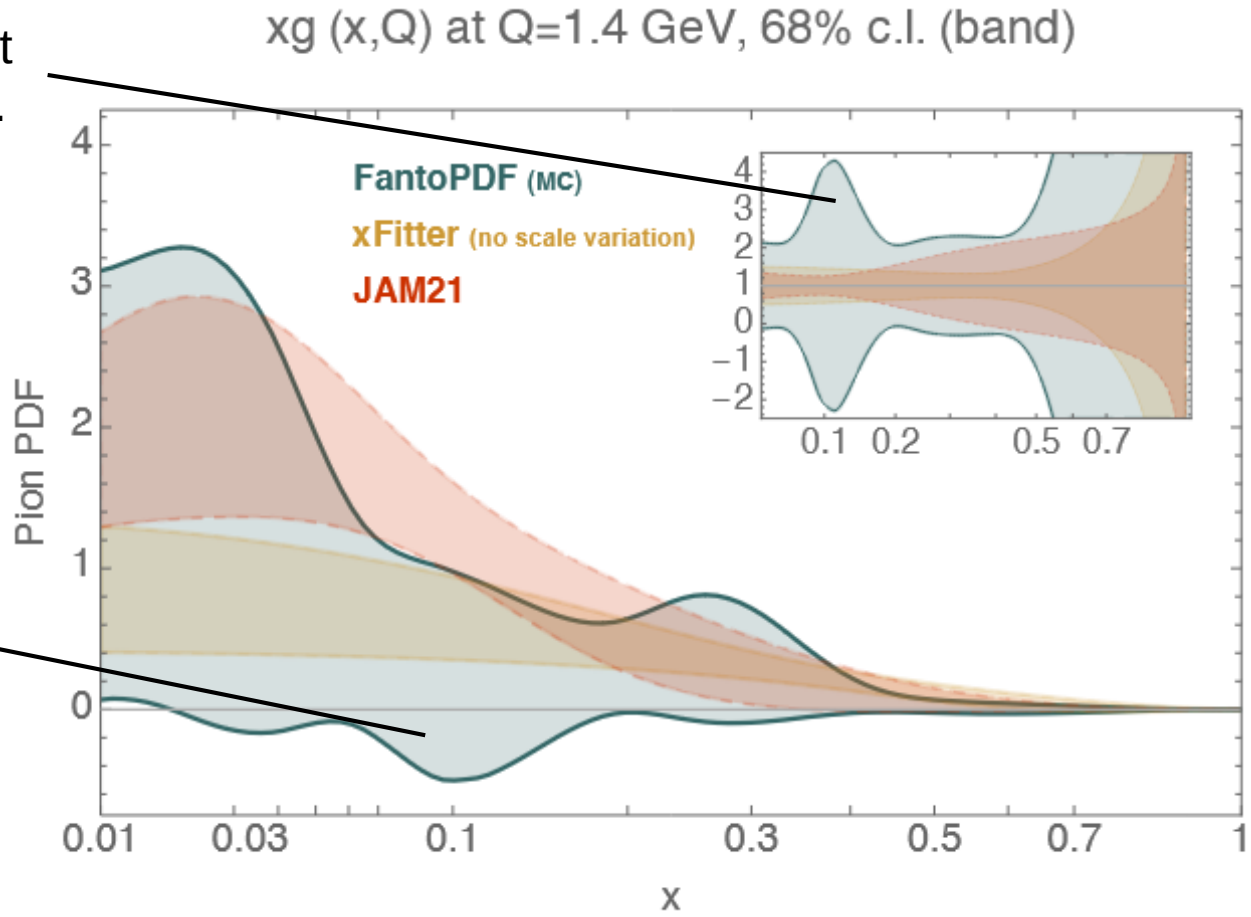
Peak is shifted to higher x.

FantoPDF has a much higher peak than JAM21 and comparative to xFitter



Larger uncertainty at extrapolation region.

Negative gluon at $x \approx 0.1$ due to extrapolation.

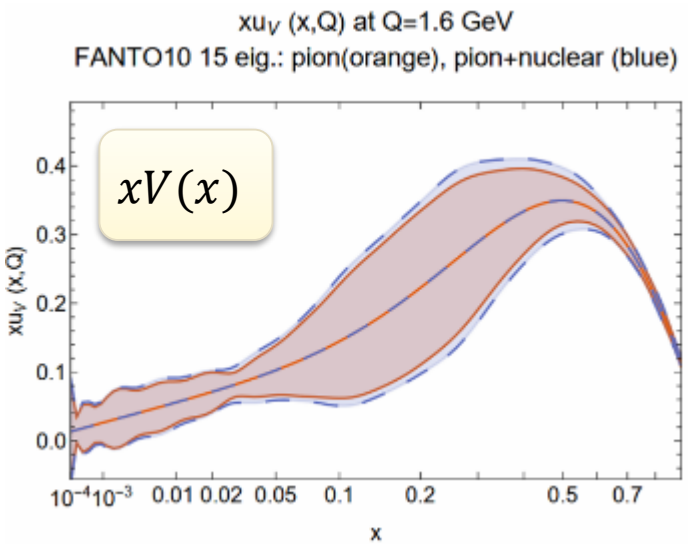


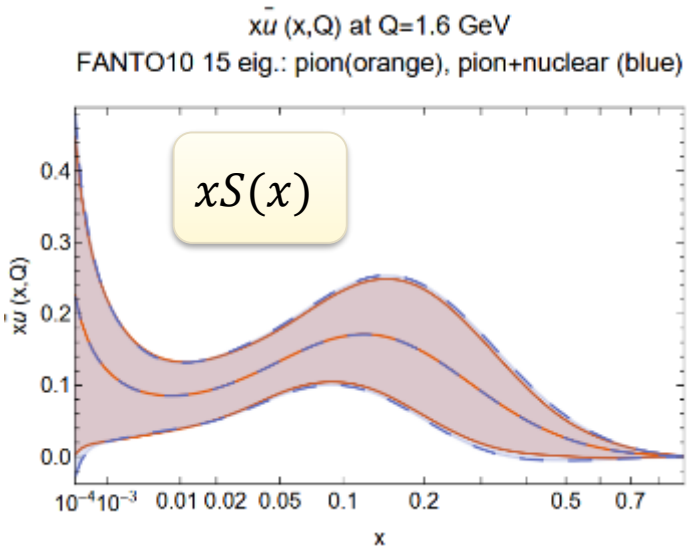
Momentum Fraction Comparison

Name	Q [GeV]	$\langle xV \rangle$	$\langle xS \rangle$	$\langle xg \rangle$
FantoPDF (DY+γ+LN)	$\sqrt{1.9}$	0.49(8)	0.34(19)	0.18(12)
xFitter [51] (DY+γ)	$\sqrt{1.9}$	0.55(6)	0.26(15)	0.19(16)
xFitter w/o scale variation	$\sqrt{1.9}$	0.55(2)	0.26(9)	0.19(9)
JAM'18 [50] (DY)	1.27	0.60(1)	0.30(5)	0.10(5)
JAM'18 [50] (DY+LN)	1.27	0.54(1)	0.16(2)	0.30(2)
JAM'21 [89] (DY+LN)	1.27	0.53(2)	0.14(4)	0.34(6)
JAM'21 [89] (DY+LN) +NLL double Mellin	1.27	0.46(3)	0.15(7)	0.40(5)
CT18 NLO (proton)	$\sqrt{1.9}$	0.443(6)	0.160(10)	0.396(10)
CT18 NNLO (proton)	$\sqrt{1.9}$	0.451(5)	0.157(10)	0.390(10)

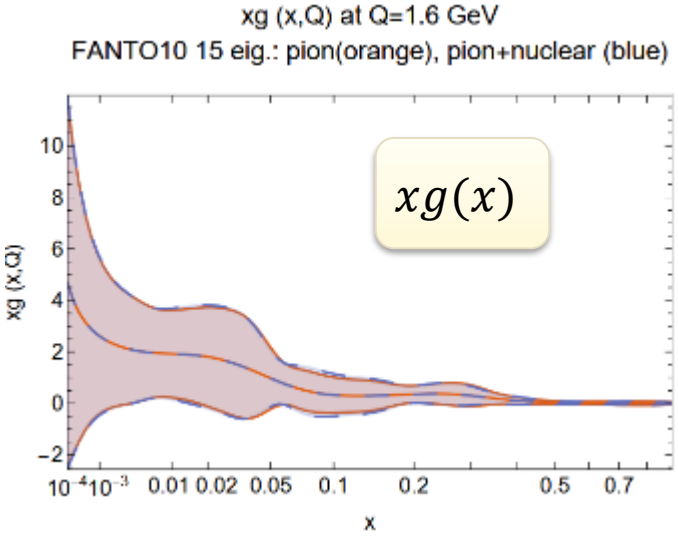
Inclusion of Nuclear PDF Uncertainties

- The DY experiments use tungsten as a fixed target.
- FantoPDF uses only the central set of the nPDF (nCTEQ15) during the fitting process, ignoring the uncertainty of the nPDF.
- 100 MC replicas were generated for the nPDF.
- The central set was fitted using each of the 100 MC nPDFs.





Final Hessian sets with model-dependent epistemic uncertainty and nuclear corrections.

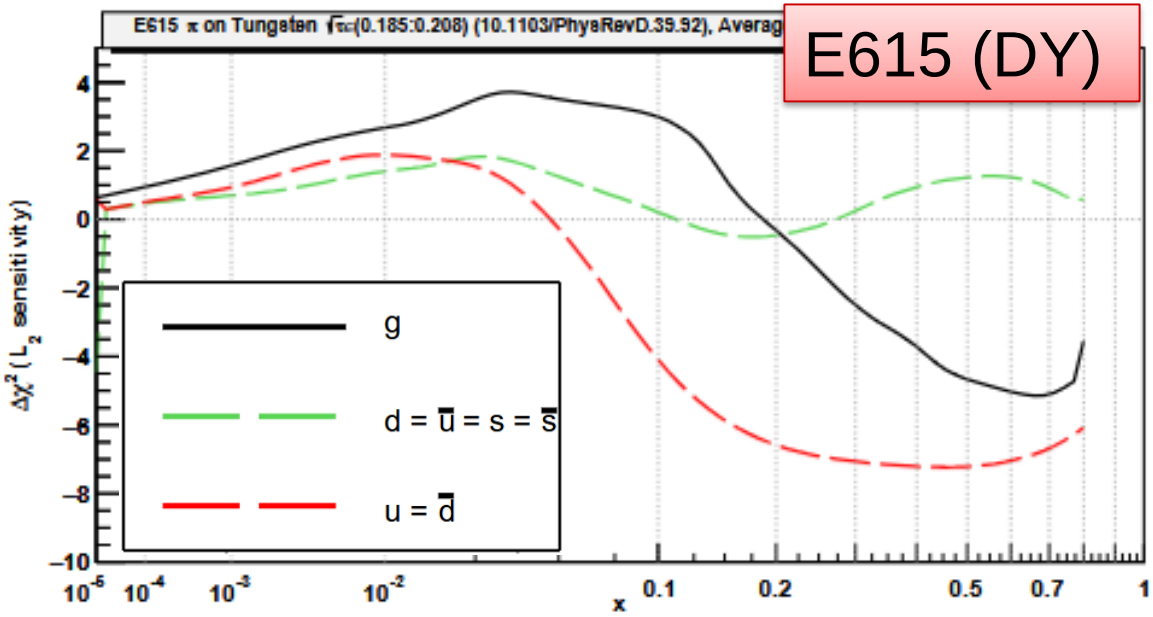


- Nuclear corrections to FantoPDF is small compared to the uncertainty from model-dependence, but non-negligible.

Exploring statistical influence of pion datasets on the pion and proton PDFs (2401.11350)

Using the L_2 sensitivity method (T.J. Hobbs et al., 2019, 1904.00022)

L_2 sensitivity averaged over five fits



L_2sensitivity is the change in χ^2 for the experiment when the indicated PDF increased by 1σ.

Positive $\Delta\chi^2$: Experiment prefers smaller PDF.

Negative $\Delta\chi^2$: Experiment prefers larger PDF.

DY data has largest influence

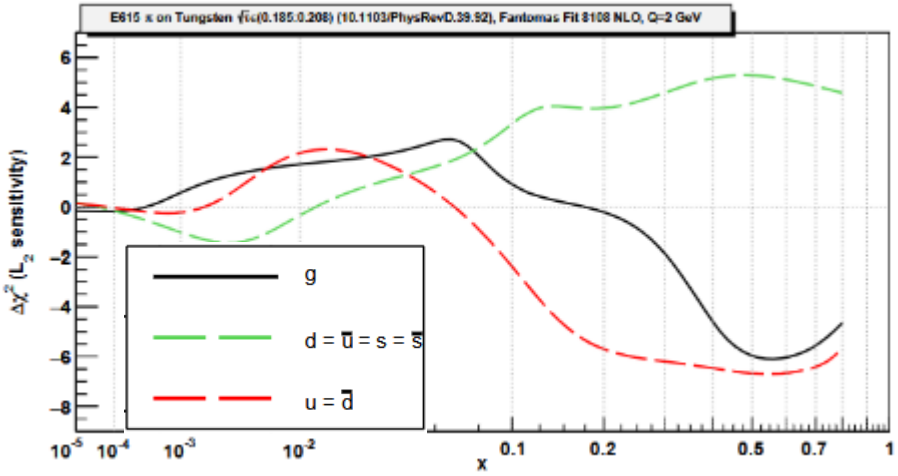
Valence: $\boldsymbol{V(x) = 2[u(x) - d(x)]}$

Quark sea: $\boldsymbol{S(x) = 6d(x)}$

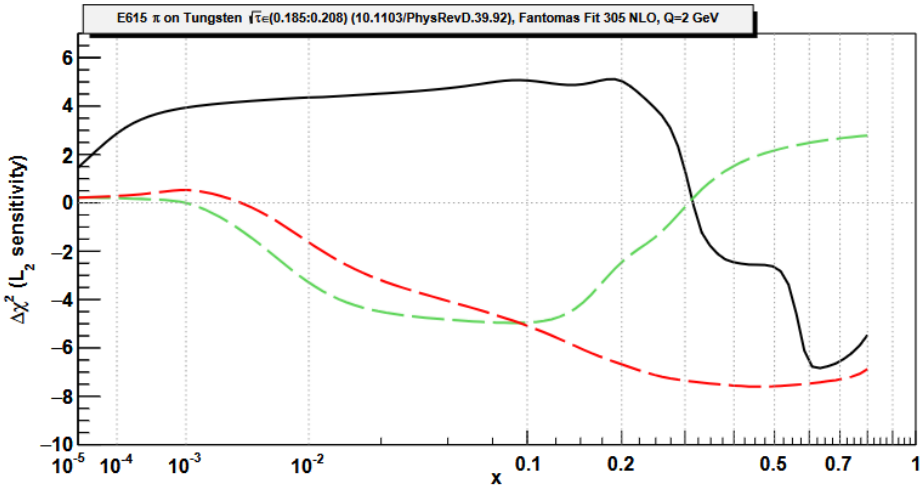
Gluon: $\boldsymbol{g(x)}$

Dataset Influence on Individual Fits

E615, $Q = 2$ GeV

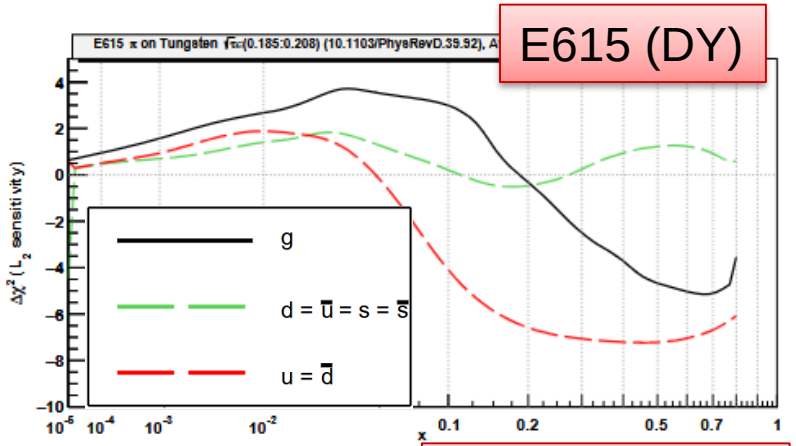


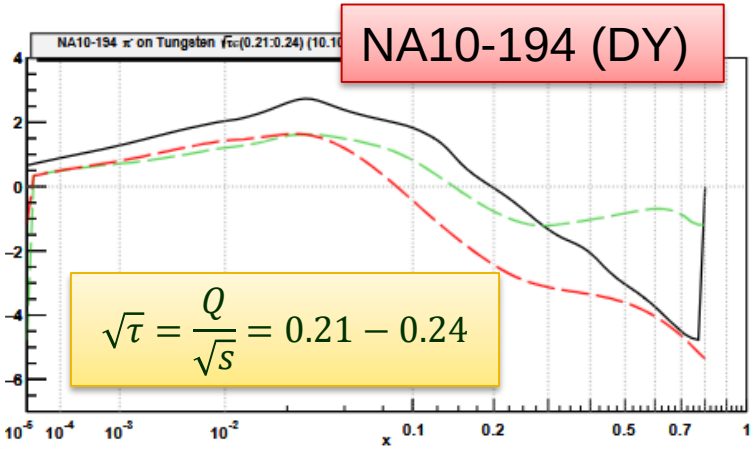
$\langle xS \rangle = 0.64$
$\langle xg \rangle = 0$



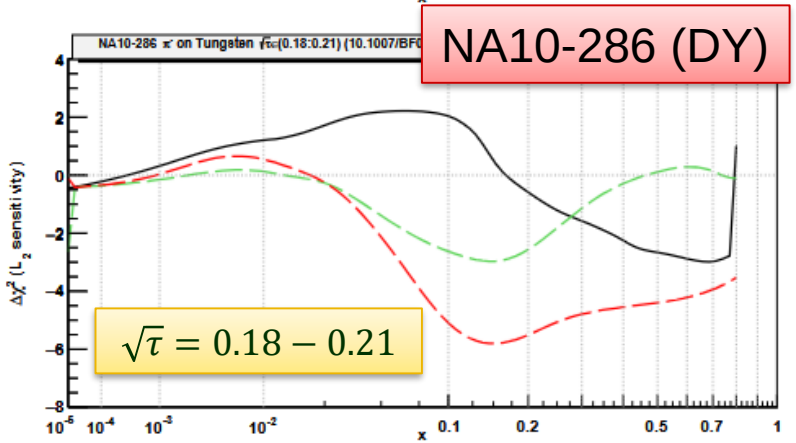
$\langle xS \rangle = 0.20$
$\langle xg \rangle = 0.28$

Average Influence on Pion Fits



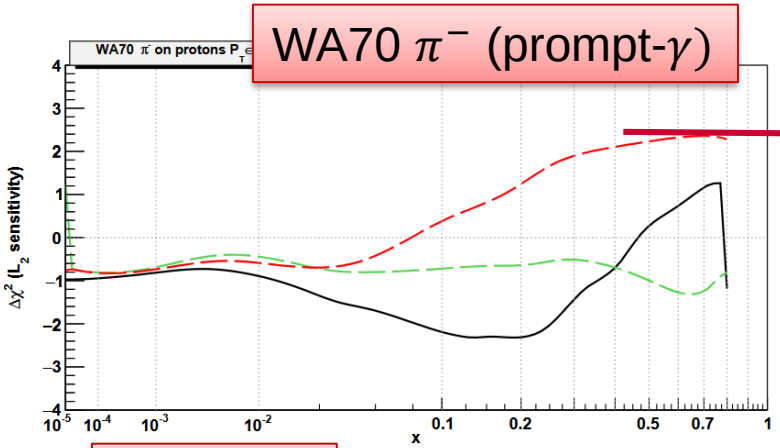


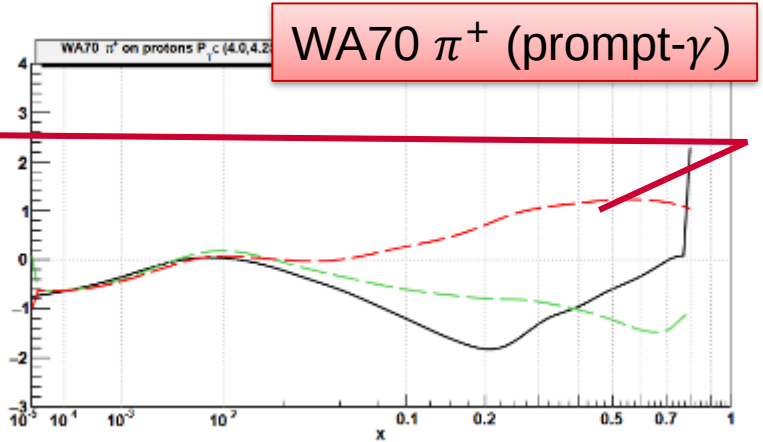
DY data has largest influence



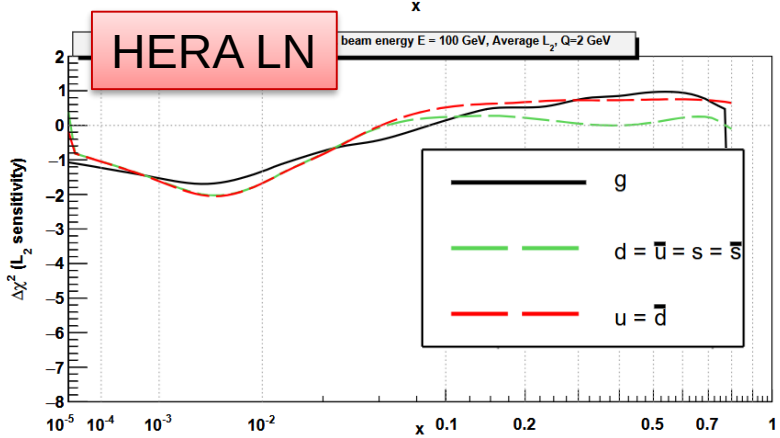
- Smaller g at low x ($x < 0.1$) and larger V at large x.
- Not very sensitive to the S PDF.
- NA10-286 ($\sqrt{\tau} = \frac{Q}{\sqrt{s}} = 0.18 - 0.21$) prefers a larger S and lower g at the extrapolation region $x \sim 0.1$.

Average Influence on Pion Fits





Prefers smaller V at large x. Opposite of DY data sets.



- Larger g at extrapolation region ($x \sim 0.1$) and Smaller V at large x for prompt-γ.
- Prompt-γ not very sensitive to the S PDF.
- Smaller g, S, and V is preferred by the LN dataset at low x ($10^{-4} < x < 10^{-2}$).

Now study the L_2 sensitivities of HERA and LHC datasets to proton PDFs (2401.11350)

L_2 calculation process using **xFitter** and **L2LHAExplorer**

Extract necessary PDF sets

Input PDFs into **xFitter**

- **CT18**
- **CT18As**
- **MSHT20**

(NNLO, $\alpha_s(M_Z) = 0.118$, $Q = 2\ \mathrm{GeV}$, $T^2 = 10$)

Extract necessary values from xFitter output

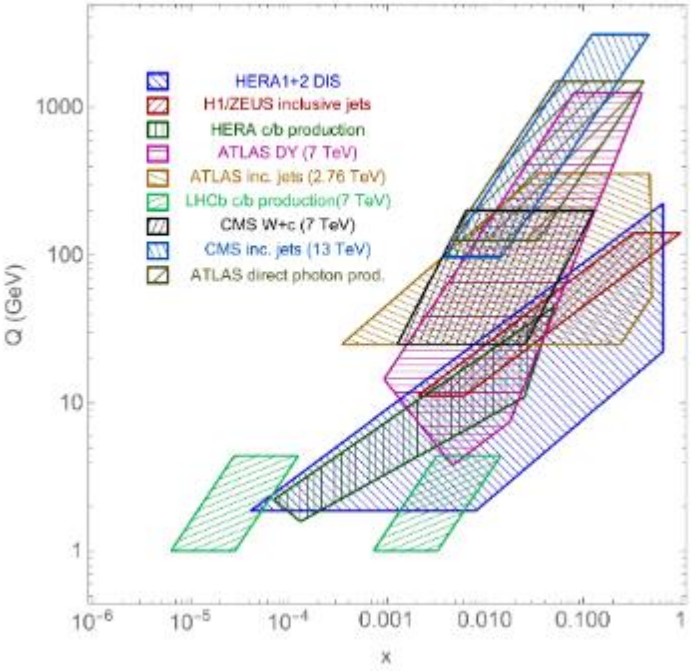
Use xFitter values to create L2LHAExplorer input

Run **L2LHAExplorer** and examine plots

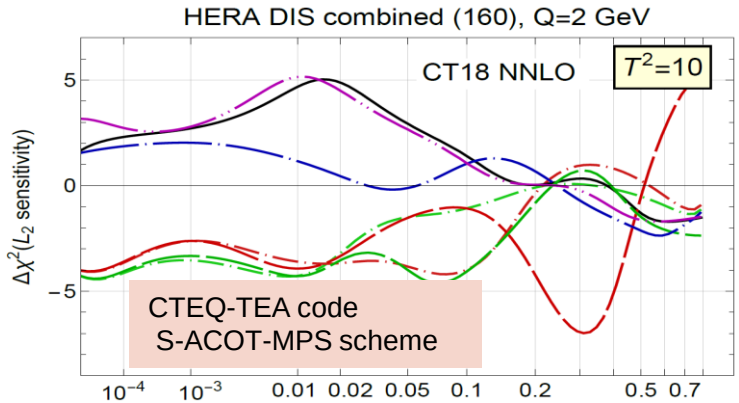
Experiments Studied

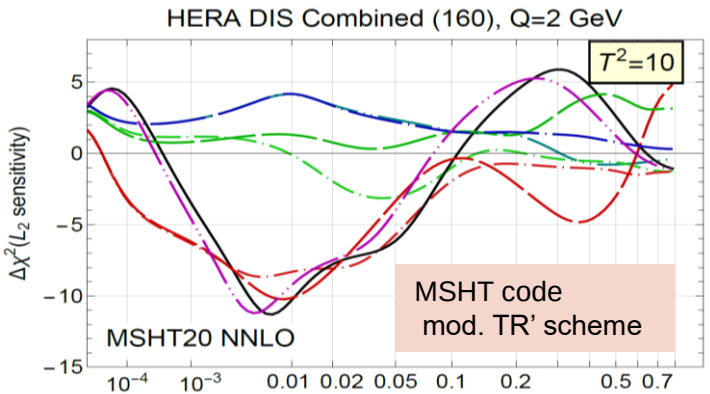
	CT18	CT18As	MSHT20
ATLAS direct γ production - 8 and 13 TeV	×	×	×
ATLAS DY - 7 TeV	×	×	✓
ATLAS inc. jet - 2.76 TeV	×	×	✓
CMS inc. jet - 13 TeV	×	×	×
CMS $W + c$ - 7 TeV	×	×	×
H1+ZEUS c and b production	×	×	✓
H1 jet			
HERA I+II DIS	✓	✓	✓
LHCb c and b – 7 TeV	×	×	×
ZEUS jet			

Approximate {x,Q} for all experiments



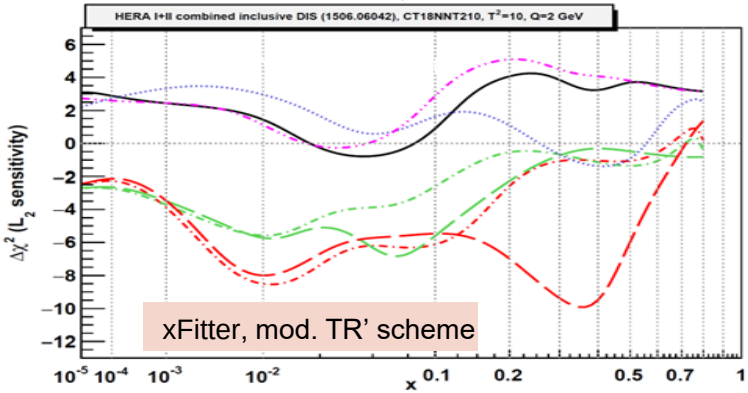
HERA I+II combined inclusive DIS [in CT18 and MSHT20]

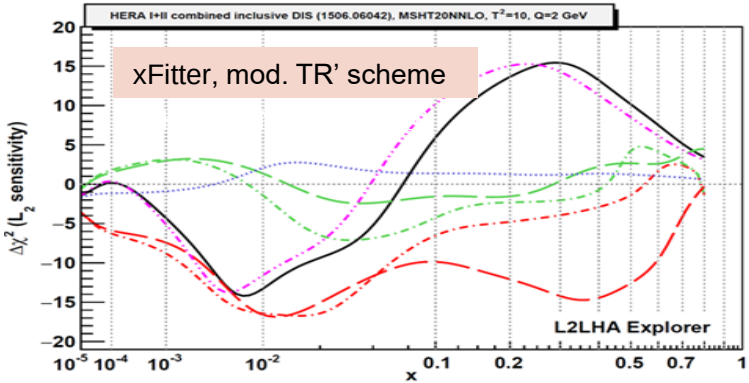




Upper row:

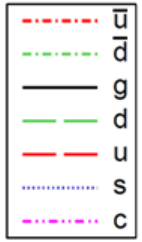
From Jing et al.,2306.03918





Lower row:

From L. Kotz, 2401.11350



Left column: differences in χ^2 definition and heavy-quark scheme. Same PDFs and m_Q.

Right column: differences in χ^2 definition only. Same PDFs and m_Q.

Conclusion & Outlook

- A wider range of possible solutions were explored with Bézier curves previous groups were unable to, leading to the quantification of model-dependent uncertainties.
- Data from the EIC will help disentangle the sea and gluon PDFs, something current data available is unable to do.
- Corrections from the nPDFs are small compared to the model-dependent uncertainties. Further studies on nPDFs may reveal dependency of nPDF choice.
- Future studies with metamorph functions and planned papers currently in the work.

Personal Contributions (2311.08447, 2401.11350, and future publications)

- Wrote C++ code for to calculate metamorphs and store multiple metamorphs with a user-defined flavor ID.
- Incorporated Fantômas module into **xFitter**'s routine and produce corresponding metamorph output files.
- Performed hundred of fits with various configurations to explore several viable pion models.
- Upgraded software used to generate MC replicas from Hessian sets **(mcgen)**.

Personal Contributions (2311.08447, 2401.11350, and future publications)

- Combined selected good pion fits into one ensemble using **mcgen** (**FantoPDF**).
- Included corrections from nuclear PDF uncertainties into FantoPDF.
- Streamlined the process to examine influences of individual datasets on any given PDF set using **xFitter** and **L2LHAExplorer**.
- Estimated the average impact of pion datasets on PDFs using the L_2 sensitivity method.

Thank you!

Contributors: Pavel Nadolsky, Fred Olness, Aurore Courtoy, Maximiliano Ponce-Chavez, Tim Hobbs (Fantômas team)

CTEQ

CONAHCYT
CONSEJO NACIONAL DE HUMANIDADES CIENCIAS Y TECNOLOGÍAS

I·AN Inter-American Network of Networks QCD

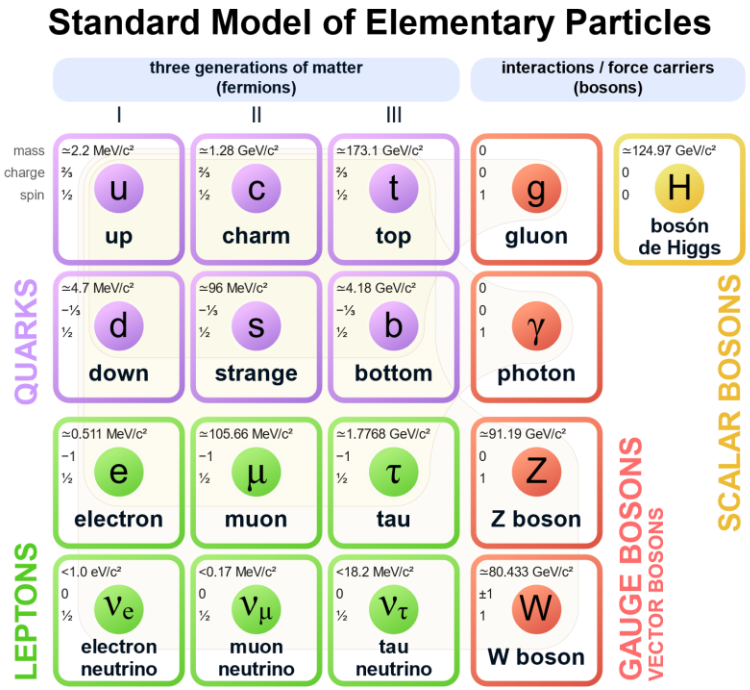
Extra Slides

Significance of the pion

- The pion is a meson, making it the lightest meson within the Standard Model ($m_{\pi^0} = 135$ MeV; $m_{\pi^\pm} = 140$ MeV).
- A mediator particle for the strong force between nucleons was theorized by Hideki Yukawa in the 1930s. It was found later that this theorized particle was the pion.
- As a result of the pion's role within QCD, it is directly responsible for generating a majority of the nuclei mass in the universe. (1907.08218)
- The pion also plays a crucial role within cosmology as it limits the energies of cosmic rays that survive collisions with the cosmic microwave background via. (Phys. Rev. Lett. 16, 748)

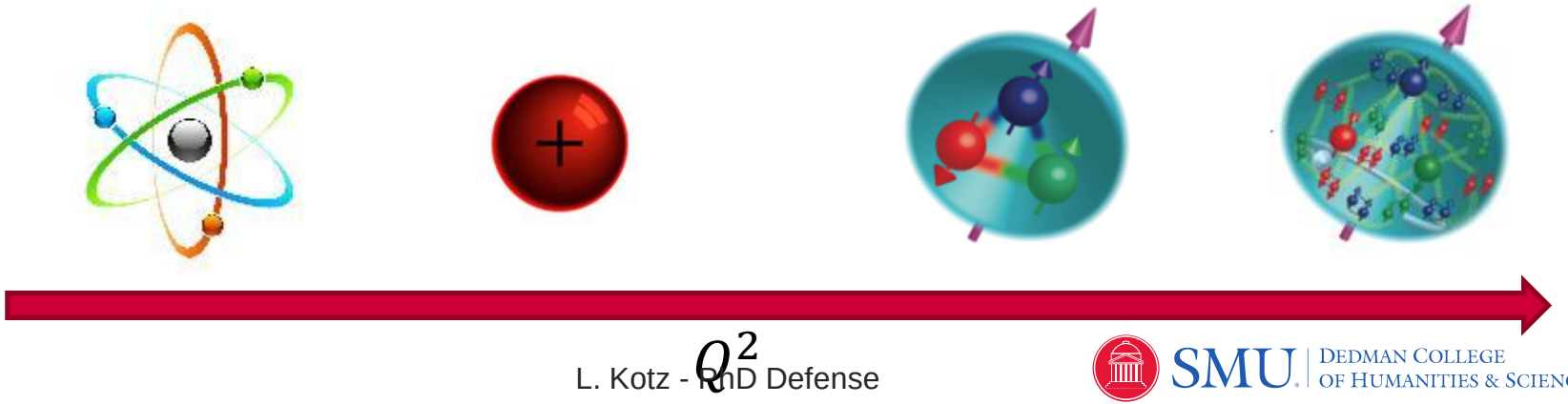
Improvements to the Standard Model

- The accuracy of various measurements at colliders are influenced by
- There are many questions unanswered pertaining to the SM:
  1. How can we achieve more accurate measurements on the SM predictions?
     - PDG quotes PDFs contributing to $\sim 50\%$ of the uncertainty on the average LHC m_W measurement.
  2. How do we know if we've detected Beyond SM signatures?
  3. What energies does the transition from non-perturbative QCD to perturbative QCD arise?



The Hadron Structure

- Hadrons are categorized by having either 3 quarks (baryons) or a quark-antiquark pair (meson). These quarks (valence quarks/non-singlet quark operator) contribute to the hadron's non-zero quantum numbers.
  - The valence quarks contribute $\sim 10\%$ of the total hadron's mass
- By increasing the energy resolution (Q^2), we find hadrons behave as a loosely bound system of partons (quarks and gluons). This described an aspect of QCD: asymptotic freedom.



Pions as a bound state within QCD

- The 2-flavor QCD free Lagrangian is:

$$\mathcal{L}_{0,n_f=2} = i\overline{\Psi}\gamma^{\mu}(\partial_{\mu} - igA_{\mu})\Psi - \mathrm{M}\overline{\Psi}\Psi; \qquad \Psi = \begin{pmatrix} u \\ d \end{pmatrix}.$$

- Assuming scales of $Q \sim \Lambda_{\mathrm{QCD}} \sim 0.2\ \mathrm{GeV}$, the mass term can be ignored as $m_u \sim m_d \ll \Lambda_{\mathrm{QCD}}$.
- The free, massless QCD Lagrangian symmetry group is $\mathrm{SU}(2)_{\mathrm{L}} \times \mathrm{SU}(2)_{\mathrm{R}} \times \mathrm{U}(1)_{\mathrm{L}} \times \mathrm{U}(1)_{\mathrm{R}}$. L and R represents the treatment of left- and right-handed fermions.
- The distinguishable symmetries between left- and right-handed fermions is known as *chiral symmetry*.

Pions as a bound state within QCD

- The transformations Ψ undergoes is:
  - $\Psi_{\mathrm{L}} \rightarrow U_L \Psi_L, ; \qquad \Psi_R \rightarrow U_R \Psi_R.$
- To examine the symmetries of the $\mathrm{SU(2)_L} \times \mathrm{SU(2)_R} \times \mathrm{U(1)_L} \times \mathrm{U(1)_R}$ group more closely, it's best to examine the group in terms of its *axial* and *vector* components ($\mathrm{SU(2)_A} \times \mathrm{SU(2)_V} \times \mathrm{U(1)_A} \times \mathrm{U(1)_V}$).
- The left- and right-handed groups can be broken down into axial and vector components for the special cases of $U_L = U_R = e^{i\alpha}\boldsymbol{I}$ and $U_L = U_R^* = e^{-i\alpha}\boldsymbol{I}$, which refer to the *axial* and *vector* symmetries respectively.

Pions as a bound state within QCD

- To examine whether the symmetries within the $\mathrm{SU(2)_A} \times \mathrm{SU(2)_V} \times \mathrm{U(1)_A} \times \mathrm{U(1)_V}$ group is preserved, the charge currents of each group is examined.
- Examining each of the groups individually reveals:
  1. $\mathrm{U(1)_A}$ contains an anomaly, $\partial_\mu j_A^\mu \neq 0$.
  2. $\mathrm{U(1)_V}$ symmetry is conserved. This results in the conserved charge, the *quark number*.
  3. $\mathrm{SU(2)_A} \times \mathrm{SU(2)_V}$ only contains one conserved charge, *isospin*.

Nambu-Goldstone bosons

- Experimental evidence shows hadrons (baryons) cannot be categorized according to a second $\mathrm{SU}(2)$ symmetry. This means there is no classification which distinguishes left- and right-handed components of hadrons. To reconcile this, the axial generators must undergo *spontaneous symmetry breaking.*
- Each broken symmetry generator results in the emergence of a *Nambu-Goldstone boson* (NGB). In the case of $\mathrm{SU}(2)$, there are three generators broken, which result in three NGBs: the three pion states.
- The mass of u and d are ignored. If we assume a non-zero mass, then the pions have a mass associated with them. Thus, pions are referred to as *pseudo-NGBs*.
- Closer analysis reveals that the pion is a composite state, which interacts with nucleons with the coupling strength, $g_{\pi NN}$ via:

$$\mathcal{L}_{\pi NN} = -i g_{\pi NN} \partial_\mu \pi^a \bar{\mathcal{N}} T^a \gamma^\mu \gamma_5 \mathcal{N}$$

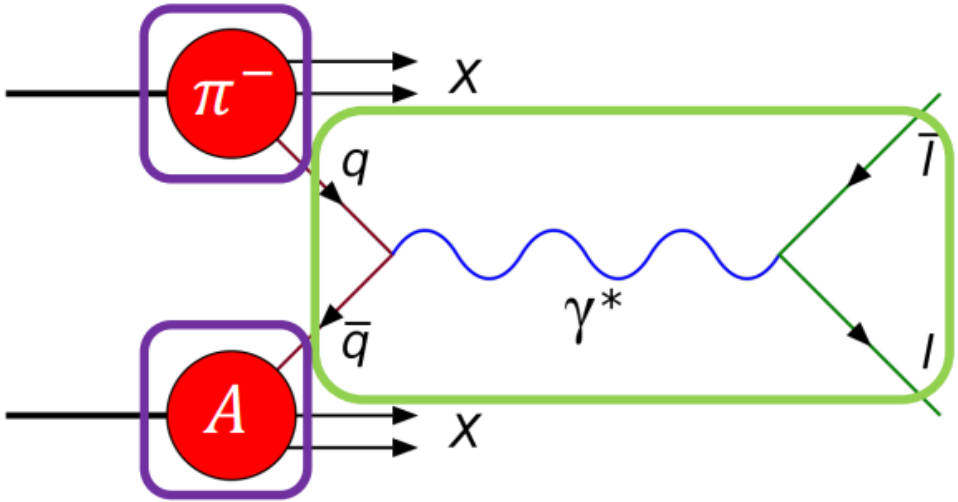
Key Properties of Our Polynomial for PDF Parameterization

1. Obey typical PDF behaviors:
   - Regge limit: $f_i(x, Q^2) \sim x^B$ as $x \to 0$; $\quad B > 0$.
   - Quark counting rules: $f_i(x, Q^2) \sim (1 - x)^C$ as $x \to 1$; $\quad C \sim 2 - 5$.
   - Sum rules are obeyed (i.e. momentum)
2. Smooth continuous function.
3. Positive-definite on the range $0 < x < 1$.
4. A polynomial function that can approximate any arbitrary, continuous function similar to neural networks ability to approximate any function via the universal approximation theorem.
5. Physically interpretable parameters.

Our Choice: Bézier Curves as an Advanced Polynomial Basis

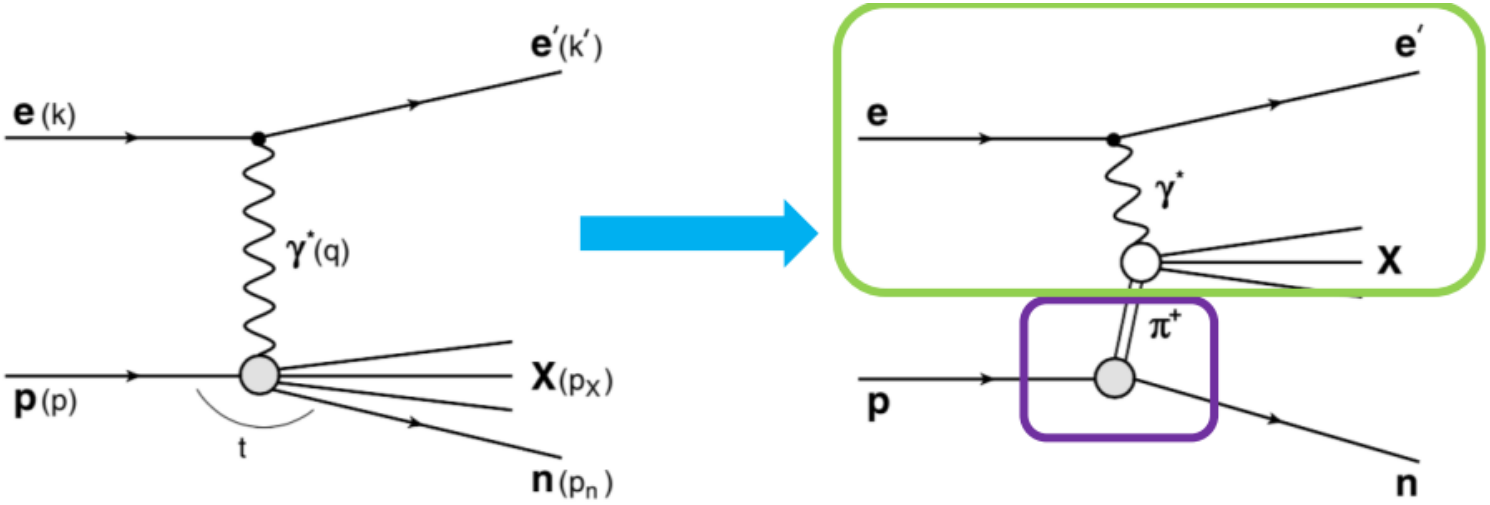
- First utilized by Paul de Casteljau and Pierre Bézier to design and manufacture the panels of the Citroën DS using CAD in the 1950s. Also commonly used in graphic design today.
- Bézier curves are a flexible polynomial, that can mimic any arbitrary function via the Weierstrass approximation theorem.
- Contain a basis of Bernstein polynomials, which are positive-definite on $[0,1]$.

Leading Order (LO) in $\alpha_s(\mu^2)$ Drell-Yan Process



$$\sigma \propto \sum_{i,j} f_i^{\pi}(x_\pi, \mu) \otimes f_j^{A}(x_A, \mu) \otimes \hat{\sigma}_{i,j}(x_\pi, x_A, Q/\mu)$$

LO Leading-neutron Process



$$\frac{d\sigma}{dx\,dQ^2\,dy} \sim f_{p\to\pi^+ n}(y) \times \sum_q \int_{x/y}^{1} \frac{d\xi}{\xi} C(\xi)\, q\left(\frac{x/y}{\xi}, \mu^2\right)$$

1

Mass-Dependent Thorne-Roberts Scheme

- The mass-dependent Thorne-Roberts (TR') scheme utilizes a general-mass variable flavor number scheme (GM-VFNS) as opposed to a fixed-flavor-number scheme (FFNS).
- When transitions to an energy scale with three active flavors $N_f = 3\ (u, d, s)$ to an energy scale with $N_f = 4\ (Q > m_c)$, there is a discontinuity within the PDFs as a function of Q.
- The TR' scheme interpolates between two schemes:
  - FFNS at low Q^2. Heavy quarks are treated only in final states.
  - Zero-mass-VFNS at high Q^2. Heavy quarks are treated as massless
- Discontinuities are avoided at $Q \sim m_{c/b}$ by matching structure functions and their derivatives.

Runge's phenomenon

- Instability of $\underline{\underline{T}}$ arises dues to vanishing determinant.
  - Equidistant spacing of control points.
  - $N_m > 5 - 10$
  - Two control points are close.
- Frobenius matrix norm is used to calculate the condition number of $\underline{\underline{T}}$
  - $\mathrm{CondNum} \propto 10^{N_m}$

Parameterizing pion PDFs

$$V(x) = 2[u(x) - d(x)]$$

$$S(x) = 6d(x)$$

$$g(x)$$

- The metamorphs parameterize the pion flavors as $xf_i(x, Q_0^2) = A_i x^{B_i}(1-x)^{C_i}[1 + \mathcal{B}^{(N_m)}(y(x))]$, where i corresponds to the independent π^+ PDFs, $q = \{g, S, V\}$.
- A_V is fixed via the valence sum rule. Either A_S or A_g is an independent parameter while obeying the momentum sum rule:

$$\langle xg \rangle + \langle xS \rangle + \langle xV \rangle = 1$$

Derivation of π^+ PDFs

- It's assumed that all sea-quarks arise from gluon splitting. As a result, all $q\bar{q}$ pairs must be counted. The quark singlet representation is used for simplicity ($q_+ = q + q^c$).

$$S = u_+ + d_+ + s_+$$

- However, the expression above also contains the valence quarks. To eliminate the valence quarks, the quark non-singlets are subtracted out ($q_V = q_- = q - q^c$).

$$\begin{aligned} S &= [u_+ + d_+ + s_+] + [-u_- - \bar{d}_-] \\ &= [u + \bar{u} + d + \bar{d} + s + \bar{s}] - u + \bar{u} - \bar{d} + d \\ &= 2\bar{u} + 2d + s + \bar{s} \end{aligned}$$

Determining the Hadron Structure

- The full cross section for any arbitrary process with two colliding partons (Drell-Yan [DY]) is given by:

$$\sigma_{\mathrm{DY}}(Q) = \sum_{a,b=g,q,\bar{q}} \int_0^1 d\xi_a \int_0^1 d\xi_b \, \hat{\sigma}\left(\frac{x_a}{\xi_a}, \frac{x_b}{\xi_b}, \frac{Q}{\mu_R}, \frac{Q}{\mu_F}; \alpha_s(\mu_R)\right) \times f_a(\xi_a, \mu_F) f_b(\xi_b, \mu_F) + \mathcal{O}\left(\frac{\Lambda_{\mathrm{QCD}}^2}{\mathrm{Q}^2}\right)$$

- Using QCD factorization can be applied to a lepton scattering off a parton as well (Deep-inelastic scattering [DIS]):

$$\sigma_{\mathrm{DIS}}(Q) = \sum_{a=g,q,\bar{q}} \int_0^1 d\xi_a \, \hat{\sigma}\left(\frac{x_a}{\xi_a}, \frac{Q}{\mu_R}, \frac{Q}{\mu_F}; \alpha_s(\mu_R)\right) \times f_a(\xi_a, \mu_F) + \mathcal{O}\left(\frac{\Lambda_{\mathrm{QCD}}^2}{\mathrm{Q}^2}\right)$$

LN datapoints

- The H1 collaboration identified the neutron's longitudinal momentum fraction range, $0.68 < x_L < 0.7$, and low momentum transfer as the kinematic region where the pion PDF can be reliable extracted from LN production (29 points).
- The pion flux function, $f_{\pi N}(x_L, t)$, gives the probability density for the proton to emit a virtual pion and become a neutron as a function of x_L, and the pions momentum, t.
- To relate the measured structure function of the LN process, F_2^{LN}, to the pion structure function F_2^{π}, the total flux of pions is required:

$$\Gamma_\pi(x_L) \equiv f_{\pi N}(x_L) = \int_{t_0(x_L, p_T^2)}^{t_{min}(x_L)} dt f_{\pi N}(x_L, t)$$

LN datapoints

$$\Gamma_\pi(x_L) \equiv f_{\pi N}(x_L) = \int_{t_0(x_L, p_T^2)}^{t_{min}(x_L)} dt f_{\pi N}(x_L, t)$$

- Consequently, the pion structure function can be proportionally represented in terms of the LN structure function and the total pion flux:

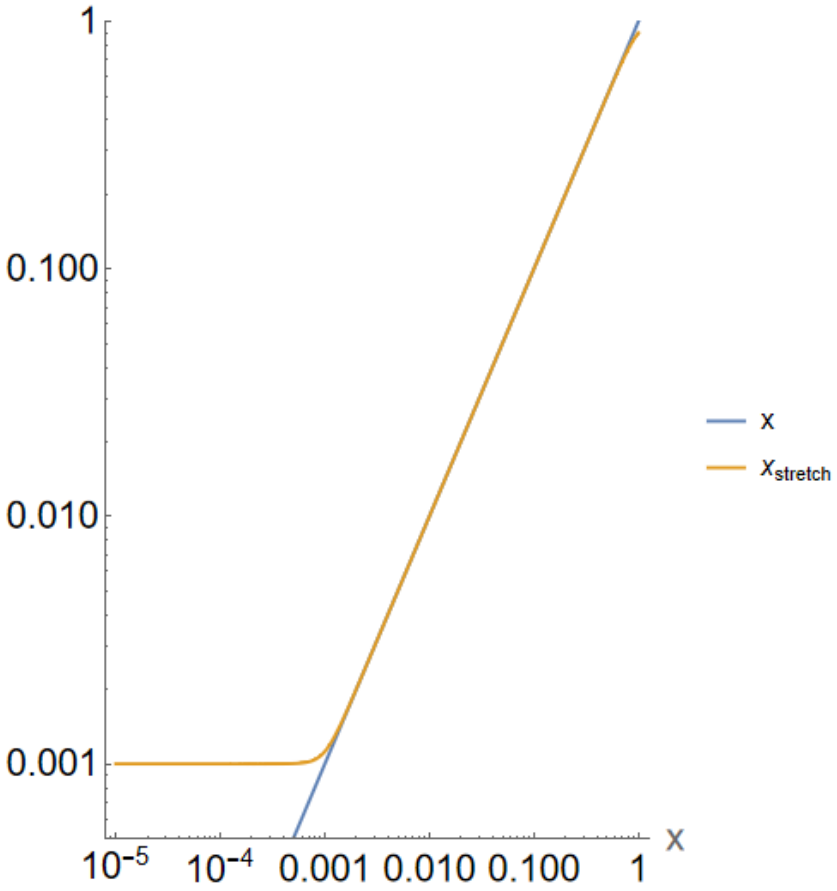
$$F_2^\pi(x_\pi, Q^2) \propto \frac{F_2^{\mathrm{LN}}(x, Q^2, x_L)}{\Gamma_\pi(x_L)}$$

- x is the Bjorken-x of the full proton system: $x = \frac{Q^2}{2p \cdot q}$
- x_π is the Bjorken-x of the pion target: $x_\pi = \frac{x}{1 - x_L}$

Stretching function

$$y(x) = g(x)^{\alpha_s}$$

$$g(x) \equiv \frac{1}{\left(\frac{1}{x^6+x_{\min}^6} + \frac{1-x_{\max}^6}{x_{\max}^6}\right)^{1/6}}.$$



xFitter parameterization

- $xV(x, Q_0^2) = A_V x^{B_V}(1-x)^{C_V}$
- $xS(x, Q_0^2) = \frac{A_S}{B(B_S+1, C_S+1)} x^{B_S}(1-x)^{C_S}$
- $xg(x, Q_0^2) = A_g\left(1 + C_g\right)(1-x)^{C_g}$
- xFitter uses the beta function $B(B_S + 1, C_S + 1)$ to set $A_S = \langle xS \rangle$.

Comparing **xFitter** pion PDFs with Fantômas module results

$$xf_i^{(0)}(x, Q^2) = A_i x^{B_i}(1-x)^{C_i}$$

$N_m = 0$ (DY+γ)	χ^2 [$d.o.f = 379 - 5$]	$\langle xV \rangle$	$\langle xS \rangle$	$\langle xg \rangle$
$B_S = 0.07$	445.70	0.556	0.268	0.177
$B_S = 0.27$	445.38	0.557	0.239	0.204
$B_S = 0.47$	445.29	0.558	0.217	0.225
$B_S = 0.67$	445.36	0.559	0.199	0.243
$B_S = 0.87$	445.52	0.559	0.184	0.257
$B_S = 1.27$	445.76	0.559	0.172	0.269

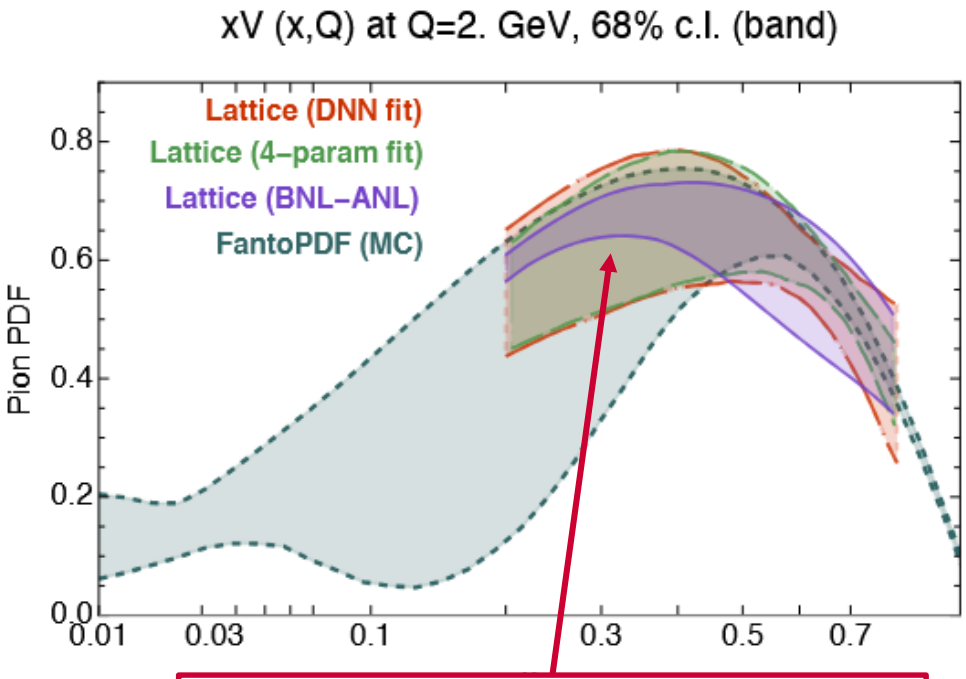
- The B_S parameter can vary considerably without affecting the statistical significance of the overall fit.
- The sea and gluon momentum fractions can invert and still produce statically similar results.

χ^2 Contribution of Each Experiment

$\langle xg(Q_0)\rangle$	$\langle xS(Q_0)\rangle$	E615 (140)	NA10-194 (67)	NA10-286 (73)	WA70 (99)	HERA F_2^π (29)	χ^2_{tot} (408)
0.32	0.14	204.7	107.7	101.5	25.6	4.7	450.1
0.27	0.17	205.5	106.4	101.5	28.8	4.3	450.8
0.20	0.28	207.5	100.7	102.5	27.0	7.1	447.6
0.12	0.45	208.4	99.6	94.9	31.6	2.1	440.4
0	0.64	204.6	95.8	93.6	29.6	4.94	432.2

Comparison Against Lattice QCD

Lattice covers xV at $0.2 < x < 0.8$



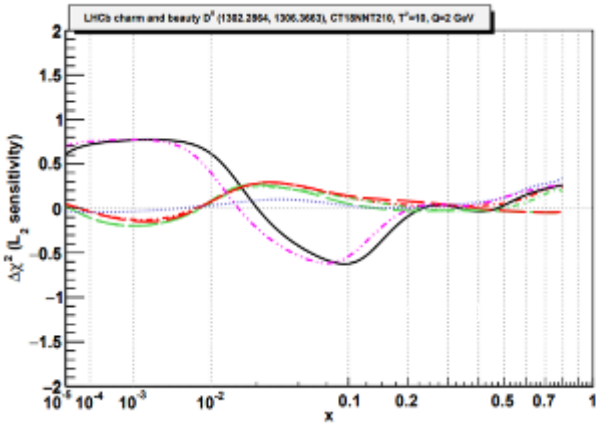
FantoPDF has larger uncertainty

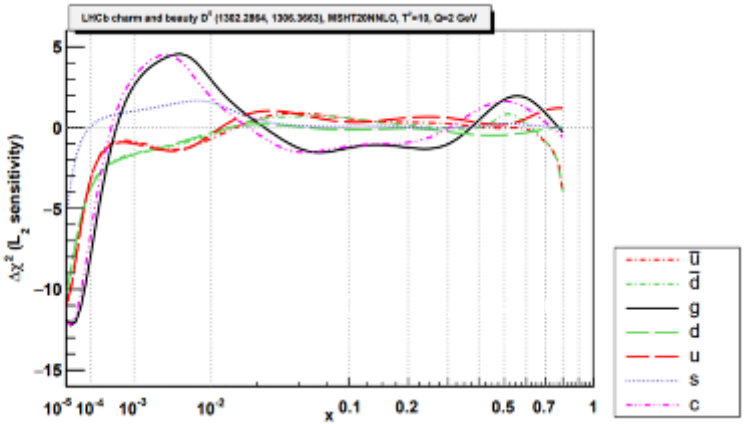
Methodology-dependent for Lattice ($0.5 \lesssim C_V^{eff} \lesssim 2$

Name	Q[GeV]	$\langle x(u+\bar{u})_{\pi^+}\rangle$	$\langle xg\rangle$
FantoPDF	2	0.331(25)	0.24(10)
HadStruct [133]	2	0.2541(33)	—
Ref. [134]	3.2	0.216(19)(8)	—
ETM [143]	2	0.261(3)(6)	—
ETM [130]	2	$0.601(28)\|_{u+d}$	0.52(11)
Ref. [144]	2	—	0.37(8)(12)
Ref. [140]	2	—	0.61(9)
Ref. [142]	2	—	0.364(38)(36)
ZeRo Coll. [145]	2	0.245(15)	—
Ref. [146]	7	0.02	—

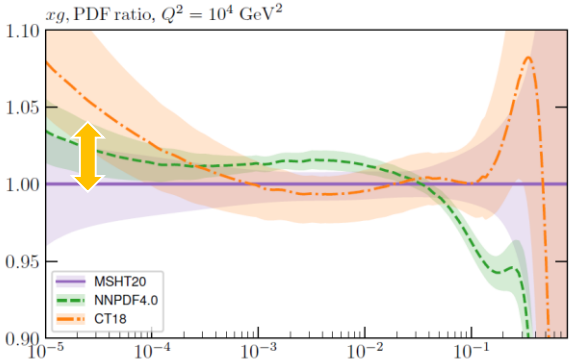
LHCb c and b @7 TeV; $p_T^{\text{meson}} \geq 2$ GeV [Not in CT18 or MSHT20]

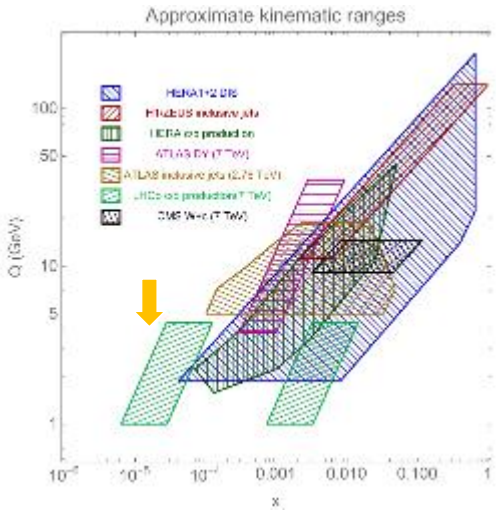
From L. Kotz, 2401.11350



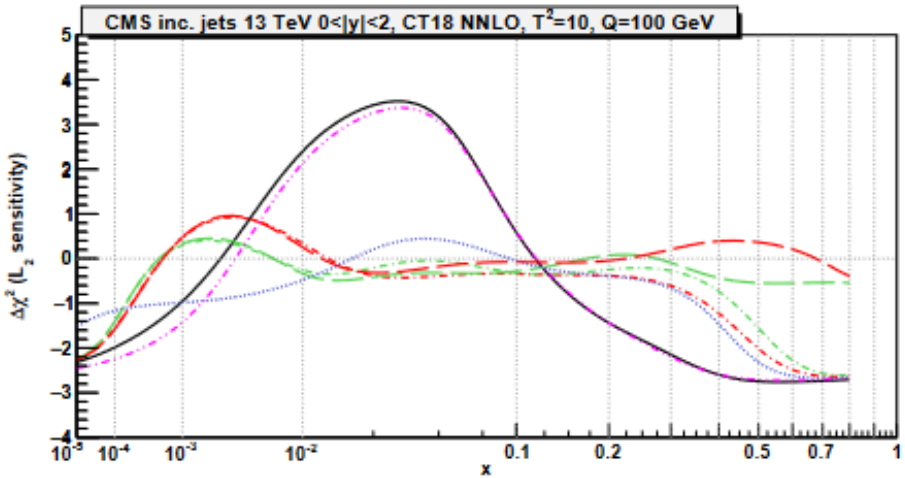


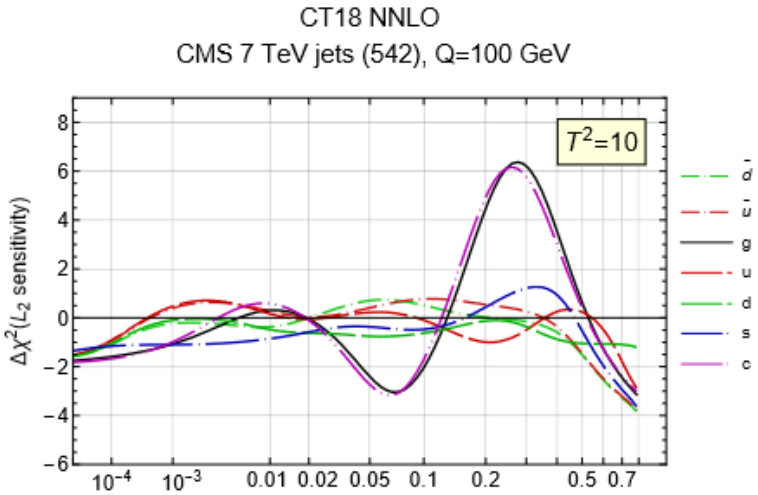
- LHCb c and b production prefers about the same gluon for CT18/CT18As.
- A larger gluon is preferred for MSHT20 at $x < 10^{-4}$.

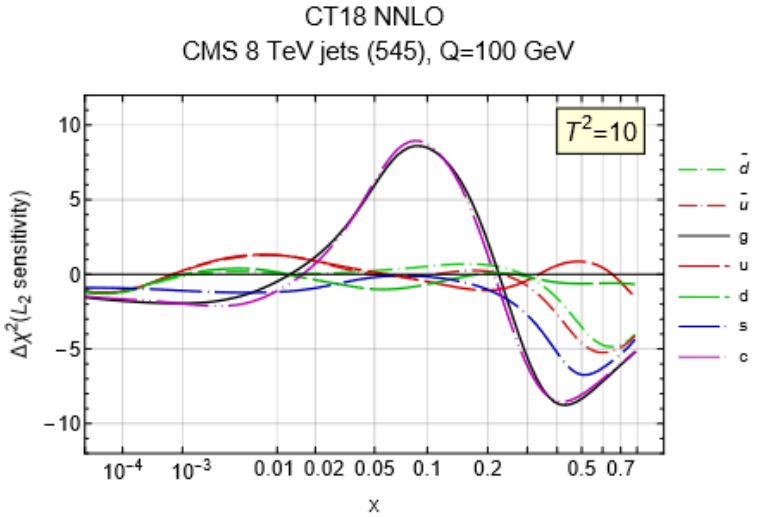




CMS inclusive jets $\sqrt{s} = 13$ TeV (CT18/CT18As)







- Expect to see the position of the L_2 peak to shift a factor of $\frac{\sqrt{s}}{13\ \mathrm{TeV}}$.
- The position of the peak for 13 TeV follows the expected shift with $\sqrt{s} = 8\ \mathrm{TeV}$. However, not for 7 TeV.
- CMS 13 TeV jet data may be more compatible with the 8 TeV data than the 7 TeV data for CT18.

Influence of Datasets on Results

- L_2 sensitivity visualizes the impact of experimental data on PDF sets.
- $S_f^{L_2}$ quantifies the correlation between the PDF uncertainty and the change in χ^2.
  - Positive $S_f^{L_2}$ results in a preferred smaller PDF.
  - Negative $S_f^{L_2}$ results in a preferred larger PDF.
- Compared my methodology using **xFitter** against CTEQ-TEA's study on various PDF sets (X. Jing et al.)

JAM Parameterization

- Chooses the template of the functional form for non-perturbative distributions are,

$$T_i(x,\mu_0^2) = \frac{N_i\, x^{a_i}(1-x)^{b_i}(1+c_i\sqrt{x}+d_i x)}{B[a_i+2,b_i+1]+c_i B[a_i+\frac{5}{2},b_i+1]+d_i B[a_i+3,b_i+1]},$$

$$c_i = d_i = 0.$$